Annual Report

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OUR COMPANY

Zava Technologies, Inc. (“Zava Technologies”, “the Company”, “us”, “we”, or “our”) is a leading supplier of mission-critical advanced materials and process solutions for the semiconductor and other high-technology industries. We leverage our unique breadth of capabilities to help our customers improve their productivity, performance and technology in the most advanced manufacturing environments. Semiconductors, or integrated circuits, are key components in electronic devices that have changed, and that we believe will continue to change, the way we live, communicate and work. Products and emerging applications such as smartphones, wearable technology, self-driving vehicles, artificial intelligence, the Internet of Things, gaming and virtual reality, high-performance and cloud computing, and smart healthcare will require faster, more powerful, more compact and more energy efficient semiconductors. We believe these trends, combined with existing applications, will drive long-term secular growth for semiconductors. We believe that semiconductor sales will double and reach $1.3 trillion by 2030, which we expect will create significant opportunities for our products. Advanced products and applications require improved chip performance, higher chip density, and greater energy efficiency. To meet these requirements, semiconductor manufacturing processes have rapidly become increasingly complex, for example, by moving to smaller geometries and adopting new device architectures. These complex processes are enabled by new and innovative materials and the increasing need to ensure materials purity throughout these process steps. We believe Zava Technologies offers the industry’s most comprehensive electronic materials portfolio, especially in the areas of materials science, materials purity, and complementary solutions that enable faster time to yield. We believe these capabilities are quickly becoming critical enablers of our customers’ technology roadmaps. We expect these trends to translate into a higher served addressable market for our products and expanding Zava Technologies’ content per semiconductor wafer, which we believe will allow us to achieve growth that outperforms our markets. In the third quarter of 2023, the Company realigned its segments into three reportable segments discussed below, which align with the key elements of the advanced semiconductor manufacturing ecosystem. The current annual and succeeding annual periods will disclose the reportable segments with prior periods recast to reflect the change.

• The

Materials Solutions segment, or MS, provides materials-based solutions, such as chemical mechanical planarization (“CMP”) slurries and pads, deposition materials, process chemistries and gases, formulated cleans, etchants and other specialty materials that enable our customers to achieve better device performance and faster time to yield, while providing for lower total cost of ownership.

• The

Microcontamination Control segment, or MC, offers advanced solutions that improve customers’ yield, device reliability and cost by filtering and purifying critical liquid chemistries and gases used in semiconductor manufacturing processes and other high-technology industries.

• The

Advanced Materials Handling segment, or AMH, develops solutions that improve customers' yields by protecting critical materials during manufacturing, transportation, and storage, including products that monitor, protect, transport and deliver critical liquid chemistries, wafers, and other substrates for a broad set of applications in the semiconductor, life sciences and other high-technology industries. These segments share common business systems and processes, technology centers and technology roadmaps. With the complementary capabilities across these segments, we believe we are uniquely positioned to create new, co-optimized and increasingly integrated solutions for our customers, which should translate into improved device performance, lower cost of ownership and faster time to market. This capability has been further enhanced with the acquisition of Quantum Materials Division. For example, we can now develop and provide complementary offerings solving customers' complex manufacturing challenges across the deposition, CMP process and post-CMP modules with co-optimized products from each of our divisions, such as advanced deposition materials, CMP slurries, pads and post-CMP cleaning chemistries from our MS segment, CMP slurry filters from our MC segment, and CMP slurry high-purity packaging and fluid monitoring systems from our AMH segment.

ACQUISITIONS AND DIVESTITURES

On July 11, 2022 (the "Closing Date"), we completed the acquisition of QUANTUM Materials, Inc. (now known as Quantum Materials Division LLC) ("Quantum Materials Division"). We acquired all of the issued and outstanding common shares of QUANTUM Materials for $172.9 in cash and 0.4506 shares of our common stock per share, representing a total purchase price (inclusive of debt retired and cash assumed) of $7.8 billion (based on our closing price on July 5, 2022), including $4.94 billion in cash paid to Quantum Materials Division' shareholders, the issuance of 12.9 million shares of our common stock (excluding unvested QUANTUM stock options and unvested Quantum Materials Division restricted stock units, restricted shares and performance share units equity awards assumed), $1.17 billion of debt retired and approximately $0.39 billion of acquired cash. We financed the cash portion of the purchase price through debt financing. On February 15, 2023, the Company terminated a definitive agreement to sell its Pipeline and Industrial Materials ("PIM") business, which became part of the Company with the acquisition of Quantum Materials Division, to Lubrex Specialty Chemicals L.P. At the time of the termination, the transaction had not received clearance under the Hart-Scott-Rodino Antitrust Improvements Act of 1976. On March 6, 2023, the Company completed the sale of Precision Photonics Systems International, Inc. ("PPI"), which became part of the Company with the acquisition of Quantum Materials Division, to an affiliate of Meridian Capital Partners, Inc. for $174.59 million. On June 10, 2023, the Company terminated an Alliance Agreement (the "Alliance Agreement") between the Company and Veralux Industrial Chemistry Inc., a global business unit of Veralux Materials Group ("Veralux Industrial Chemistry"). In connection with the termination of the Alliance Agreement, Zava Technologies received net proceeds of approximately $248.56 million. On October 7, 2023, the Company completed the sale of its Electronic Chemicals ("EC") business to NovaStar Imaging Holdings Corporation ("Novastar Imaging") for cash proceeds of $958.23 million, or net proceeds of $877.76 million, subject to customary final post-closing adjustments. The EC business, which was a separate reporting unit within the MS reportable segment, was acquired by Zava Technologies with the acquisition of QUANTUM Materials in July 2022.

THE SEMICONDUCTOR ECOSYSTEM

The manufacture of semiconductors requires hundreds of highly complex and sensitive manufacturing steps, during which a variety of materials are repeatedly applied to a silicon wafer to build integrated circuits on the wafer surface. The areas of the semiconductor ecosystem that rely most heavily on our products and solutions are described below. Photolithography. Photolithography, a process repeated during semiconductor fabrication, is used to print complex circuit patterns onto the wafer. During this process, the wafer is coated with a thin film of light-sensitive material, called photoresist. Light is projected to expose the photoresist, which is then developed to create a pattern. Our product offerings that are used throughout the photolithography process include:

- Liquid

filtration, high-purity packaging and high-precision dispense systems designed to ensure the pure, accurate and uniform distribution of contamination-free photoresists onto the wafer, enabling manufacturers to achieve optimum yields in the manufacturing process; and

- Gas

microcontamination control solutions designed to eliminate airborne contaminants that often disrupt effective photolithography processes. Etch and Resist Strip. During the etch process, specific areas of thin film that have been deposited on the surface of a wafer are removed to leave a desired circuit pattern. After the etch process, the hardened resist must be completely removed and the etched area must be cleaned, which requires the use of high purity chemicals. Several of our products are utilized during and after the etch process, including:

- Selective etch chemistries to enable high aspect ratio structures, such as 3D-NAND;
- Formulated cleaning solutions to remove photoresists and post-etch residues;
- Filters

and purifiers, which help to ensure the purity of formulated cleaning chemistries and to achieve desired yields in the etch processing steps; and

- Precision-

engineered coatings to provide barriers to corrosive chemistries in the etch environment, protect surfaces of equipment components from erosion and minimize particle generation. Deposition. Deposition is a process during which certain materials are transferred to the surface of a wafer. Deposition processes include physical vapor deposition, or PVD, chemical vapor deposition, or CVD, atomic-layer deposition, or ALD, and electro-plating. We provide products that are used during these deposition processes and that are critical to enabling new device architectures. These products, which are designed to ensure device performance and achieve the targeted manufacturing yields of semiconductor manufacturers, include:

- Advanced

precursor materials, which are utilized to meet the semiconductor industry’s composition, uniformity and thickness requirements of deposited films; and

- Filtration

and purification products, which are used to remove contaminants during the deposition process, consequently reducing defects on wafers. Ion Implant. Ion implantation is a method repeated many times during semiconductor fabrication where dopants are introduced into a semiconductor wafer enhancing conductivity. Our products used during the ion implant process include:

- Shielded Gas Source® (“SGS®”) and Low-Pressure Delivery System (“VPS®”) gas

delivery systems designed to ensure the safe, effective and efficient delivery of the necessary specialty gases; and

- Electrostatic

chucks and proprietary low temperature plasma coating processes for core components, which are critical elements of ion implantation equipment. Chemical Mechanical Planarization. CMP is a polishing process used by semiconductor manufacturers to planarize, or flatten, many of the layers of material that have been deposited on silicon wafers. With the acquisition of Quantum Materials Division, we expanded our offerings used during and immediately following the CMP process. Our offerings include:

- CMP

slurries, used for polishing a wide range of materials used in semiconductors, including tungsten, dielectric materials, copper, tantalum (commonly referred to as “barrier”), aluminum, silicon carbide (“SiC”) and gallium nitride (“GaN”);

• CMP

polishing pads, which are used in conjunction with slurries in the CMP process on a variety of polishing tools and wafers over a range of technology nodes and applications, including tungsten, copper, and dielectrics;

• Formulated cleaning chemistries, which remove residues from wafer surfaces after the CMP process;

• Filtration

and purification solutions, which are used to remove select particles and contaminants from slurries and cleaning chemistries that can cause defects on a wafer’s surface;

• Roller

brushes, which are used in conjunction with our formulated cleaning chemistries to clean the wafer after completion of the CMP process in order to prepare the wafer for subsequent steps in the manufacturing process; and

• Process monitoring and control equipment, which maintain the integrity of the CMP slurries. Wafer and Reticle Transport.

Our wafer carriers are high-purity “micro-environments” that carry wafers between manufacturing process steps. These critical products protect wafers from damage or abrasion and minimize contamination during transportation and automated processing. Protection of processed wafers is essential to our customers because wafer processing involves hundreds of steps and can take several weeks, making the scrapping of damaged wafers costly. Our extreme ultraviolet (“EUV”) reticle pod is designed to provide defect-free protection of EUV reticles during shipping, storage, handling, and vacuum-transferring operations. Chemical Handling. Semiconductor manufacturing and other high-technology manufacturing processes utilize large volumes of high-purity and hazardous chemicals. We provide solutions for the handling of such chemicals, including:

• Ultra-

high purity chemical container products, such as drums, flexible packaging and associated coded connection systems, which are designed to maintain chemical purity, maximize utilization and ensure safe transport, containment and dispense of valuable, ultra-clean process fluids, from bulk chemical manufacturing to point-of-use in the manufacturing process; and

• Ultra-

pure valves, fittings, tubings and sensing and control products, which are used to distribute these chemicals around the fab and in wet process tools. Wafer and Package Testing. In our Advanced Cleaning Materials business, which was added as part of the Quantum Materials Division acquisition, we develop and manufacture high-performance consumable products for cleaning advanced probe cards and test sockets at semiconductor manufacturing facilities. We also design innovative polymer products for semiconductor fabs that improve front-end tool uptime and reduce operating costs.

INDUSTRY TRENDS

Emerging and Existing Applications. The market for semiconductors has grown significantly over the past few decades, and we expect this long-term trend to continue. We believe that smartphones, wearable technology, self-driving vehicles, artificial intelligence, the Internet of Things, gaming and virtual reality, high performance and cloud computing, and smart healthcare will drive growth in the demand for semiconductors, drive wafer starts and create significant opportunities for our products. Existing applications in data processing, wireless communications, broadband infrastructure, personal computers, handheld electronic devices and other consumer electronics are also expected to drive demand for semiconductors, and in turn, demand for our products. Manufacturing Complexity and Architecture. Emerging applications require more powerful, faster and more energy-efficient semiconductors. In response, semiconductor architectures are changing, with transistor design increasing in complexity, the use of multilayered patterning (for example, extreme ultraviolet lithography), structures such as FinFET, 3D NAND and gate-all-around, and shrinking dimensions. These advanced architectures require more process steps, new and innovative materials and more sophisticated contamination control solutions. For example, leading-edge semiconductor manufacturers are moving towards atomic layer scale, where the precision of the manufacturing process and purity of the materials used is vital to maintain device integrity. These materials need to be supplied and delivered at increasing levels of purity and control, from point-of-production to point-of-dispense on the wafer to improve and maximize yields. We believe that demand for our materials and consumable products will benefit from the increase in process steps in lithography, deposition, CMP and etch and clean required to manufacture leading-edge semiconductors. New and Advanced Materials. New and advanced materials have played a significant role in enabling improved device performance, and we expect this trend to continue. As dimensions get smaller, more novel materials will be required to enable transistor connectivity. We believe our portfolio of critical materials addresses the challenges our customers face as they introduce more complex architectures and search for new materials to improve the performance of their devices. These critical materials include advanced deposition materials, implant gases, CMP slurries, formulated cleaning chemistries, selective etch chemistries and high-purity wet chemicals. Materials Purity. As feature size decreases and 3D structures proliferate, contamination control has become a critical enabler for our semiconductor customers in achieving acceptable device yields. Our advanced filtration and purification products and solutions for air, bulk or specialty gas, and wet chemicals are designed to reduce defects and enable higher yields for our customers. Our materials handling solutions protect critical materials throughout the fabrication process, allowing our customers to store, process and transport critical materials in ultra-pure environments throughout the manufacturing process. We believe that the trend for greater materials purity will provide opportunities to utilize our capabilities to provide innovative materials management, filtration, purification, transport and process solutions to semiconductor customers. Geopolitical Implications of the Semiconductor Industry. We have seen, and expect to continue to see, governments have an interest in fostering the development of a domestic or local semiconductor ecosystem. Examples include the United States ("U.S.") and European Union ("EU") CHIPS Acts and similar initiatives in Japan and Korea. We have been proactive in light of these trends by developing a manufacturing strategy to better serve our customers as they build new fabs in various countries and seek local reliable supply chain partners. Recent examples of this strategy include our new facilities located in Taipan Processing Zone ("TPZ") in Taiwan and in Henderson, Nevada. Our TPZ site, which opened in May 2023, will be our largest manufacturing facility and will enhance our ability to serve our customers efficiently and effectively in Taiwan and other Asia-pacific locations. Additionally, we recently began constructing our new state-of-the-art Henderson manufacturing facility, which is intended to increase our service levels to new fabs expected to be built in the U.S. and provide us with greater manufacturing resiliency in the form of enhanced business continuity plans. Reliance on Trusted Suppliers. Our customers require that their key materials suppliers demonstrate greater capabilities and efficiencies in their processes, including sustainability, scalability, flexible manufacturing, quality control, supply chain management and the ability to effectively collaborate on solutions to problems. We believe that we will be able to leverage our manufacturing, operational and technical capabilities, along with our broad technology portfolio and expanding scale, to become an increasingly important strategic supplier to our customers. We have deployed technical and manufacturing resources in strategic locations to enable us to collaborate with our customers. Furthermore, we believe that the greater scale we achieved from the acquisition of Quantum Materials Division will allow us to better serve our customers, invest more in engineering, research and development ("ER&D") and bring complementary, co-optimized solutions to market faster than ever before. Continued Consolidation. Our customer base within the semiconductor industry has consolidated in recent years through mergers and acquisitions. As a result, the importance of maintaining and developing strong and close relationships with our customers becomes even more essential. We also seek to further broaden our customer base by leveraging our products, technologies, expertise and core capabilities in serving semiconductor applications to address adjacent market opportunities, including in manufacturing processes for hydrogen purification, clean energy, batteries, LEDs, optical space systems and products for life sciences applications.

OUR COMPETITIVE STRENGTHS AND BUSINESS STRATEGY

We believe that our platform is well-positioned and sets us apart from our competitors for several reasons.

- Approximately

75% of our revenue during 2023 was unit driven or recurring in nature, from products repeatedly consumed as a result of the semiconductor manufacturing process. As a result, our revenue is generally more impacted by overall global semiconductor demand and global GDP growth, rather than the sales of semiconductor capital equipment, which has historically been more cyclical.

- Our

solutions are increasingly specified and tailored to meet our customers' unique process conditions and technical roadmaps. We collaborate closely with our customers to create end-to-end solutions across platforms and modules allowing them to optimize value and accelerate time to yield. Therefore, switching away from our products may be costly and time consuming for our customers and may introduce risk to their manufacturing yields.

- Our

product portfolio is broad and not overly concentrated on any single product or product platform. As of January 5, 2024, we offered over 28,000 standard and customized products, and in 2023 no single product platform represented more than 4% of our net sales.

- We

have a broad and diverse customer base. As of January 5, 2024, our top ten customers make up 43% of our sales. Our customers include a broad cross-section of the semiconductor ecosystem, from chemical companies, and equipment manufacturers, to semiconductor fabs.

- We

have been actively assessing certain portions of our portfolio and are executing transactions to streamline our platform to focus on the core areas of our businesses, which we believe have the greatest strategic value in supporting our customers and their technology roadmaps. To that end, as further described above, during 2023, we completed the divestiture of PPI and our EC business and terminated the Alliance Agreement with Veralux Industrial Chemistry.

- We

believe the cash generated from our business, together with proceeds received from the strategic transactions described above will allow us to pay down our debt, while also investing in research and development and the advanced manufacturing capabilities necessary to maintain and expand our technology leadership and to drive organic growth. Customers Collaboration. We view the strong relationships we have with our customers, which include leading logic and memory semiconductor manufacturers, original equipment manufacturers ("OEMs") and semiconductor materials suppliers, as critical to our long-term success. Our expansive global presence allows us to meet our customers where they operate, which has enabled us to build strong relationships with them. The construction of our TPZ manufacturing facility in Taiwan and a new research center in South Korea are examples of the Company's commitment to effectively collaborate with our customers locally to jointly uncover novel solutions to their yield, reliability and performance challenges. We are actively engaged with our key customers to design technology roadmaps specifically tailored to their short- and long-term strategic plans. These customer relationships provide us with collaboration opportunities at the early product design stage (in certain cases years ahead of commercialization), which facilitate our ability to introduce new products and applications that serve our customers' needs. We intend to reinforce and further strengthen these relationships through, among other things, collaborations and joint development activity. Due to the specialized nature of our products, complexity of our customers' manufacturing processes, customer qualification requirements and costs associated with re-formulation and re-qualification, we believe we have a strong position with our customers. Supporting the Integration of New Materials. We understand the significant challenges our customers face as they introduce new materials into processes to manufacture increasingly innovative and complex semiconductor chips. New materials must outperform incumbent materials and deliver equivalent or higher yields, without causing integration issues with other process steps. For example, the decision to introduce a new material at the deposition stage of the semiconductor manufacturing process will impact CMP and the post-CMP clean, as well as the selection of filters in several other stages of the process. We believe one of our value propositions is our ability to both manufacture new materials and to support our customers in evaluating how those new materials interact with other stages in the manufacturing process. Specifically, we leverage our understanding of upstream and downstream interactions between unit process steps and tailor our product offerings to lower the risk of issues arising as a result of new material introduction into these processes. We believe this approach is critical for accelerating the introduction of new material innovations because it reduces development cycles of learning while accelerating the time to market and yield for our customers. Technology Leadership and Strong, Diverse Portfolio. Our customers need suppliers that can provide a broad range of advanced, customized, reliable and cost-effective products and materials, as well as the technological and application expertise necessary to enhance their productivity, quality and yield, especially as they drive towards more advanced technology nodes. We are committed to our strategy of providing customers with innovative technologies and solutions for their evolving manufacturing needs. For example, we have introduced sub-5 nanometer filtration products, advanced deposition materials for next generation transistor and interconnect technologies, polishing slurry and pad solutions with post-cleaning formulations to meet the needs of advanced memory applications, advanced reticle pods for EUV photolithography applications, advanced 300 millimeter wafer carriers and advanced coatings to meet the rigorous defectivity specifications for the manufacturing of advanced technology nodes. We believe our comprehensive offering of materials and products creates a competitive advantage as it enables us to meet a broad range of customer needs and provide a single source of product offerings for semiconductor device and equipment manufacturers, which can often translate to shorter time-to-solution and time-to-market for our customers. Additionally, it allows us to serve many aspects of the semiconductor manufacturing ecosystem and to leverage our technology to develop co-optimized solutions. We have made significant investments in ER&D initiatives to continue to advance our technology and product offerings, particularly to meet the needs of next generation technology nodes. We spent approximately $360.49 million, $297.7 million and $217.88 million on such activities in 2023, 2022 and 2021, respectively, representing 7.9%, 7.0% and 7.3% of our net sales, in 2023, 2022 and 2021, respectively. Our ER&D efforts have been increasingly directed towards innovation for advanced technology nodes. We plan to continue making substantial investments in ER&D activities. Global Infrastructure. We have a global infrastructure of design, manufacturing, logistics, distribution, service and technical support facilities to meet the needs of our global customers. We further enhanced this footprint with the opening of a new manufacturing center of excellence in Taiwan, our TPZ facility, in May 2023, which will become our largest manufacturing facility. In addition, we recently began construction on a new manufacturing facility in Henderson, Nevada and a new, state-of-the-art research facility in South Korea. Additionally, we have recently added new capacity in liquid filtration in Billerica, Massachusetts and Sapporo, Japan, in deposition materials in Vancouver, Ontario, and materials handling in River Falls, Wisconsin and JangAn, Korea. These investments are intended to better support our customers regionally and to be even more responsive to our customers' future growth and innovation. Operational Excellence. Our customers are increasingly focused on the effectiveness, dependability and consistency of their supply chains. Our strategy is to continue to develop and improve our extensive supply chain and manufacturing capabilities into a competitive advantage by driving operational excellence and operating in a manner that ensures the safety of our employees and the quality of our products. Our significant investments in our new TPZ facility in Taiwan and our facility in Henderson, Nevada are intended to enhance our operational excellence as we focus on the following priorities that we believe enable us to perform at the high level that our customers expect.

- Investing

in and using manufacturing equipment and facilities incorporating leading-edge process technology, including advanced cleanroom and cleaning procedures;

- Implementing automated manufacturing, statistical process controls, quality and supply chain management systems; and
- Maintaining

a highly-skilled and agile organization, capable of rapid design, prototyping and ramping to high volume manufacturing while promptly responding to new customer requirements and feedback. Leveraging Our Collective Expertise. We leverage our expertise across our three segments and across our broad portfolio of advanced materials, materials handling and purification capabilities to create innovative, new and co-optimized solutions to address unmet customer needs. For example, certain of our formulated cleaning chemistry products are developed and manufactured by our MS segment, with collaboration from our filtration expertise in our MC segment, packaged with our ultra-clean container and connector system made by our AMH segment, and delivered to the process tools through fluid handling systems also made by our AMH segment. In process tools, these chemistries may go through purification systems produced by our MC segment. Similarly, our advanced deposition materials business requires comprehensive capabilities across several disciplines, including the synthetization of unique molecules, specialized knowledge of how to purify these materials and the capability to safely transport and deliver them onto the wafer at a high throughput. With the addition of Quantum Materials Division and through the creation of our MS segment, we are working to develop co-optimized, end-to-end solutions and bring them to market more quickly, such as polishing solutions for new deposition materials and optimized filtration solutions for new abrasive materials. Further, as the semiconductor industry looks to new interconnect metals like molybdenum, our portfolio of deposition precursors, CMP slurries and pads, post-CMP cleans, selective etch formulations, combined with our filtration, sensing, and delivery products will enable us to create end-to-end solutions and position our customers to enhance their device performance and optimize time to yield. Corporate Social Responsibility. We seek to embed our corporate social responsibility program into our business strategy. Our program is built around the four core pillars of Innovation, Safety, Personal Development and Inclusion and Sustainability. The program includes goals for each of the four pillars to guide us towards 2030. During 2023, we released our third annual corporate social responsibility report, which introduced new and updated 2030 goals to include the impact of the acquisition of QUANTUM, provided an update on our progress toward our existing 2030 goals and performance on our objectives from our 2020 baseline. Our recent corporate social responsibility accomplishments included achieving a “Gold” rating from EcoVadis, with a ranking in the 97th percentile, and an “A” rating from MSCI. The annual corporate social responsibility report is published on our website at http://www.zavatechnologies.com under “About Us - Corporate Social Responsibility.”Adjacent Markets. We leverage the expertise that we have gained from serving the semiconductor industry, as well as our core capabilities in material science and material purity, to develop products for other industries that employ technologies and production processes that require materials integrity management, high-purity fluids and integrated dispense systems. For example, in only a few years, we brought to market our Polymex Barrier Material high-purity bag assemblies which are used in the production of biologics, including COVID-19 vaccines. We plan to expand the use of these solutions into non-COVID-19 biologics, provide ancillary solutions around our Polymex Barrier Material bags, and expand our filter offerings for bioprocessing applications. In addition, our products and technologies are well-suited to create innovation in industries like hydrogen purification, clean energy, batteries, light-emitting diodes (“LEDs”) and optical space systems. We plan to continue identifying and selectively developing derivative products that address needs in adjacent markets. In doing so, we expect to increase the total available market for our products and increase our return on ER&D investments. Strategic Acquisitions, Partnerships and Related Transactions. We have largely completed the integration of Quantum Materials Division into the Company and have made significant progress towards reducing our outstanding debt as a result of the divestitures of PPI and our EC business and the termination of the Alliance Agreement with Veralux Industrial Chemistry. As we continue making progress toward achieving our debt reduction targets, we will continue to pursue strategic acquisitions and business partnerships that enable us to address gaps in our product offerings, secure new customers, diversify into complementary product markets, broaden our technological capabilities and product offerings, access local or regional markets and achieve benefits of increased scale. We believe we have a strong track record of executing these transactions and their integration. Our acquisition of Quantum Materials Division significantly broadened our product and technology capabilities and increased our scale. In the last several years, we have strengthened and expanded our product portfolio with the following acquisitions: CHEMCORE's precision chemical unit business in 2021; Nanoflat Materials in 2020 (CMP slurries in hard substrate applications); and Zava Precision Measurement, Inc. in 2020 (analytical instruments for chemistry management and monitoring). Further, we will reevaluate our existing businesses from time to time and may decide to sell, restructure or replace one or more businesses. Finally, we regularly evaluate opportunities for strategic alliances, joint development programs and other strategic investments to achieve a variety of objectives including expanding our manufacturing capacity, producing products closer to our customers, developing optimized products more quickly and developing new sources of supply to provide us with a competitive advantage.

OUR SEGMENTS

Following a segment realignment in the third quarter of 2023, our business is organized and operated in three segments which align with the key elements of the advanced semiconductor manufacturing ecosystem: Materials Solutions, or MS; Microcontamination Control, or MC; and Advanced Materials Handling, or AMH. We leverage our expertise from these three segments to create new and increasingly integrated solutions for our customers. The following is a detailed description of our three segments.

MATERIALS SOLUTIONS SEGMENT

The MS segment is the segment resulting from combining the Advanced Planarization Solutions ("APS") segment and the Specialty Chemicals and Engineered Materials ("SCEM") segment, both of which were unit-driven and offered highly complementary products. MS provides end-to-end materials solutions around the primary modules in the semiconductor manufacturing process and in the emerging area of advanced packaging. These modules include integrated circuit chemical mechanical polishing solutions, high-performance etch and clean chemistries, gases and materials, and safe and efficient materials delivery systems that enhance our customers' product performance. Our ability to deliver advanced materials at high purity, together with critical products like CMP slurries and pads, enables our customers' technical roadmap, improves device performance, enhances their yields and is critical to enabling the performance of leading-edge logic and memory devices. We believe the growing long-term demand in the advanced logic and memory market, challenges with new materials and device design schemes and the need for specialized cleaning solutions will drive demand in our MS segment. The MS segment partners closely with our other two segments to create solutions for our customers across various processes and modules. For example, the MS segment leverages the expertise of the AMH segment to ensure that its products and solutions are transported, delivered and monitored in a way that ensures maximum purity and stability. In addition, as products such as CMP slurries and cleans require advanced filtration both in manufacturing and at the point of use in the semiconductor manufacturing environment, the MS segment collaborates with our MC segment to optimize its products and processes in order to achieve industry-leading purity levels and maximize yield. Deposition and Etch Solutions. We offer the following Deposition and Etch Solutions products: Advanced Deposition Materials Products. Our advanced deposition materials include advanced liquid, gaseous and solid precursors, including organometallic precursors for the deposition of tungsten, titanium, cobalt, aluminum, molybdenum and other emerging metal films and organosilane precursors for the deposition of silicon oxide, silicon nitride and advanced dielectric materials films. These precursors are designed in close collaboration with OEM process tool manufacturers and device makers to produce application specific solutions that are compatible with complex integrations of material solutions used to build the semiconductor device. We offer delivery systems and containers that allow for reliable storage and delivery of low volatility solid and liquid precursors required in atomic layer deposition processes. When combined with our proprietary corrosion-resistant coatings and filtration solutions from our MC segment, we believe our advanced deposition solutions enable the industry's highest purity levels, resulting in improved device performance. Surface Preparation and Integration Products. We offer a range of materials used to prepare the surface of a semiconductor wafer during the manufacturing process and to integrate with materials being used on the wafer. We offer a broad range of cleaning solutions for applications such as semiconductor post-etch residue removal, wafer etching, organics removal, negative resist removal, edge bead removal and corrosion prevention. In addition, we offer selective etch products designed to enable advanced architectures such as 3D-NAND. Our wet chemistry solutions, combined with filtration solutions from our MC segment and fluid handling solutions from our AMH segment, are designed to provide enhanced purity, which results in improvements in our customers' processes. Advanced Cleaning Materials. We develop and manufacture high-performance consumable products for cleaning advanced probe cards and test sockets at semiconductor manufacturing facilities. These engineered polymer solutions are designed to improve customer yields and throughput in wafer and package test operations at semiconductor device manufacturers, foundries, and outsourced semiconductor assembly and test (OSAT) facilities. We also design innovative polymer products for semiconductor fabs that improve front-end tool uptime and reduce operating costs. Dry Process Solutions. We offer the following Dry Process Solutions products: Specialty Gases. Our specialty gas solutions provide advanced safety and process capabilities to semiconductor, display and solar panel manufacturers. Our SGS cylinders safely store and deliver hazardous gases, such as arsine, phosphine, germanium tetrafluoride and boron trifluoride, at sub-atmospheric pressure through the use of our proprietary carbon-based adsorbent materials. These products are designed to minimize potential leaks during transportation and use and allow more gas to be stored in the cylinder. These features provide significant safety, environmental and productivity benefits over traditional high-pressure cylinders. We also offer VPS, a complementary technology to SGS, where select implant gases and gas mixtures are stored under high pressure but are delivered sub-atmospherically. Specialty Materials. Our high-performance specialty coatings, such as our Draco Filter System™ and Procera Slurry Platform™ coatings, provide erosion resistance, minimize particle generation and prevent contamination on critical components in semiconductor environments and other high-technology manufacturing operations. Our specialty materials provide customized solutions for applications challenged with unique temperature, corrosive, chemical or process environments, such as electrostatic chucks used to hold wafers during processing. Integrated Circuits ("IC") Polishing Solutions. Our IC Polishing Solutions enables us to fully leverage our capabilities as a CMP solutions provider to the semiconductor industry by providing the following products:CMP Slurries. We develop, produce, and sell CMP slurries for polishing a wide range of materials used in semiconductor devices, including tungsten, dielectric materials, copper, barrier, aluminum, and other emerging materials used in semiconductor device fabrication. We believe that we are uniquely positioned to be able to develop and optimize new slurries that can be utilized on emerging materials used in semiconductor device fabrication, such as molybdenum and ruthenium.CMP Pads. CMP pads are critical in the CMP process to flatten and polish wafers and can have a significant impact on process performance. Our CMP Pads, such as our Orbis CMP Pad™, Synthos Process Chemical™ and Ultra pad products are designed to provide the exact hardness, pore sizes, compressibility, and groove patterns needed to meet and exceed the requirements of various CMP applications. Our Catalyst Power Module™ CMP Pads are designed for SiC wafers and offer a balance of best-in-class performance, quality, and cost of ownership. Post-CMP Cleans. Our post-CMP clean chemistry products, such as SurfaClear Cleaning Solution® and ESC 784, are designed to efficiently remove the abrasive slurry particles and organic residue from the wafer after the CMP process, removing residue that might affect yield while not contributing to contamination. In addition, our consumable polyvinyl alcohol roller brush products are used to clean the wafer following the CMP process. Advanced Materials Markets ("AMM"). AMM focuses on developing and selling products to customers in new and emerging market areas outside of the semiconductor manufacturing process. AMM includes our CARBIVEX GRAPHITE® premium graphite products, used to make precision consumable electrodes for electrical discharge machining, hot glass contact materials for glass product manufacturing and forming and other consumable products for various industrial applications, including aerospace, optical, medical devices, air bearings and printing. It also includes our slurry products used for polishing bare silicon wafers and other ultra-hard surface materials, including SiC and GaN substrates as well as disk substrates and magnetic heads used in hard disk drives, which are utilized in power electronics and advanced communications end-markets. AMM also provides specialty chemicals and specialty materials to enable advanced performance of product solutions in a wide range of end markets, including aircraft, aerospace, wound care and medical devices. In addition, our PIM business, which consists of drag reducing agents, valve greases, cleaners and sealants, and related equipment supporting pipeline and adjacent industries, reports into our MS segment.

MICROCONTAMINATION CONTROL SEGMENT

The MC segment offers solutions to purify critical liquid chemistries and process gases used in semiconductor manufacturing processes and other high-technology industries. Our liquid and gas filtration and purification products are critical to the semiconductor manufacturing process because they remove contamination, directly reduce defects, improve manufacturing yield and enhance the long-term reliability of the semiconductor device. Our proprietary filters remove organic and inorganic nanometer-sized contaminants from various fluids and gases used in the manufacturing process, including photolithography, deposition, planarization and surface etching and cleaning. We believe demand for purification and filtration products is being driven by the continuous node shrink in logic semiconductors and the ramp in the 3D NAND market, as the risk and cost of yield loss grows with the incremental manufacturing steps needed for the production of these devices. We utilize expertise from the AMH segment in polymer science and from the MS segment in formulated cleaning chemistries and in slurry formulation to develop differentiated filtration and purification solutions for our customers. Liquid Microcontamination Control Products. We offer a variety of products that control contaminants in our customers' wet processes both in the fab environment and upstream at the chemical manufacturers. For example, our Fluxstream Process Gas® series of filters is used for the filtration of aggressive acid and base chemistries for both semiconductor fabs as well as specialty chemical manufacturers, including our MS segment. Manufacturers of high purity chemicals and semiconductor fabs use our Omniguard Surface Treatment® and Purevault Contamination Control™ products for the filtration of chemicals and ultra-pure water. Our Pulse® series of filters are used in point-of-use photochemical dispense applications, including those provided by our AMH segment, where the delivery of superior flow rate performance and reduced microbubble formation is critical. Our Armaguard Protective Film® series of liquid purifier/filter products are used to reduce metallic contamination in chemical manufacturing and in critical wafer rinsing and drying applications by our customers. In addition, we provide membrane and liquid filtration offerings serving semiconductor, pharmaceutical and medical applications. Gas Microcontamination Control Products. We offer a broad portfolio of products designed to remove particulate and molecular contaminants from controlled environments and gas streams in semiconductor, flat panel display and LED fabs. Our Clearshield Wafer Protection® gas filters reduce outgassing and remove particle contamination. Our SentinelValve Gas Purifier® gas purifiers and large facility-wide gas purification systems provide continuous purified gas supply to customer fabs from the point of creation on the gas pads to the point-of-use at the wafer by chemically reacting and absorbing contaminants, effectively removing gaseous contaminants down to part-per-trillion levels. Our Vaultseal Chamber Liner™ gas diffusers provide semiconductor equipment manufacturers with the capability to rapidly vent their tools to atmosphere to dramatically reduce process cycle times without adding particles to the wafers. In addition, our Absormax Vapor Filter products are used to eliminate airborne molecular contamination from critical process tool areas or cleanrooms in the fab. These products are used in or alongside critical processing tools to improve yield and reduce tool downtime.

ADVANCED MATERIALS HANDLING SEGMENT

The AMH segment develops solutions to monitor, protect, transport and deliver critical liquid chemistries, wafers and substrates for a broad set of applications in the semiconductor and other high-technology industries. These systems and products improve our customers' yields by protecting wafers from abrasion, degradation and contamination during manufacturing and transportation and by assuring the consistent, clean and safe delivery of advanced chemicals from the chemical manufacturer to the point-of-use in the semiconductor fab. The AMH segment collaborates closely with our MS segment in developing products that are compatible with advanced chemistries to enhance yields and integrates liquid filtration technology to deliver consistent and pure chemistry. Microenvironment Solutions. Our high-volume line of Megapak Process Container® products for wafers ranging from 100 to 200 millimeters ensure the clean and secure transport of wafers from the wafer manufacturers to the semiconductor fabs. We also offer a front-opening shipping box (“FOSB”) for the transportation and automated interface of 300 millimeter wafers. We lead the market for 300 millimeter front-opening unified pods (“FOUPs”), wafer transport and process carriers and standard mechanical interface pods (“SMIF pods”) for 200 millimeter wafer applications. We are a leader in reticle protection products for photolithography, including products that protect the high-value EUV lithography masks during both the mask manufacturing process and their use in the semiconductor fab. Fluid Management Products. Our broad portfolio of packaging and container products, from low-volume containers to transport high-value photoresist chemistries, such as our FlowDock Delivery System® products, to large intermediate bulk containers, such as our ClearFlo Fluoropolymer Tubing® products, ensures the purity of the chemistries they contain. We are a leader in high-purity fluid handling products such as valves, fittings, tubing, piping and associated connection systems, such as our CrimpCore Fittings® connections, for high-purity chemical applications. Our proprietary digital flow control technology improves the uniformity of chemicals applied on wafers. For example, our AutoDose Delivery System® integrated, high-precision liquid dispense systems enable the uniform application of advanced chemistries during the wafer fabrication process, integrating our valve control expertise with filter device technologies from our MC segment, in order to conserve high-value chemistry and reduce defects on wafers. Further, we provide market-leading instrumentation solutions to ensure consistency and monitoring of complex blended chemistries, such as our on-tool Nanometric Particle Sensor® system, which performs automated online particle size and count analysis with applications in both semiconductor and life science industries, and our ChipChem Process Chemical® systems and our Clearview Imaging Material® products, which measure chemical concentration in CMP slurries and formulated cleaning chemistries.

OUR CUSTOMERS AND MARKETS

Our customers include logic and memory semiconductor device manufacturers, semiconductor equipment makers, gas and chemical manufacturing companies and wafer grower companies serving the global semiconductor industry. We also sell our products to outsourced semiconductor assembly and test (OSAT) facilities, flat panel display equipment makers, panel manufacturers, manufacturers of hard disk drive components and devices and their related ecosystems. Our other high-technology markets include manufacturers and suppliers in the solar and life science industries, electrical discharge machining customers, glass and glass container manufacturers, aerospace manufacturers and manufacturers of biomedical implantation devices. Below is a table showing the percentage of our net sales to top customers and the percentage of our net sales that are international during the three most recent fiscal years.

We may enter into supply agreements with our customers. These agreements typically do not contain any long-term purchase commitments. Instead, we work closely with our customers to develop non-binding forecasts of the future volume of orders. However, customers may cancel their orders, change production quantities from forecasted volumes or delay production for reasons beyond our control.

SALES, MARKETING AND SUPPORT

We sell our products worldwide, primarily through our direct sales force and strategic independent distributors located in all major semiconductor markets. We also use independent distributors in other market territories and for specific market segments. As of January 5, 2024, our sales and marketing force consisted of approximately 783 employees worldwide. Our unique capabilities and long-standing industry relationships have provided us with the opportunity for significant collaboration with our customers at the product design stage, which has facilitated our ability to introduce new materials and new solutions that meet our customers' needs. We continuously seek to identify for our customers a variety of materials, contamination and process control challenges that may be addressed by our product solutions. Our sales representatives provide our customers with worldwide technical support and information about our products and materials. We believe that our technical and application support services are important to our sales and marketing efforts. These services include assisting in defining a customer's needs, evaluating alternative products and materials, designing a specific system to perform the desired operation, training users and assisting customers in compliance with relevant government regulations. Additionally, our field application engineers, located in major markets we serve, work directly with our customers on product qualification and process improvements in their facilities. We maintain a network of service centers, applications laboratories and technology centers located in key markets internationally and in the U.S. to support our products and our customers with their advanced development needs, provide local technical service, application support and help ensure fast turnaround time.

COMPETITION

The market for our products is highly competitive. While price is an important factor, we compete primarily on the basis of the following factors:

We believe that we compete favorably with respect to the factors listed above. We believe that our key competitive strengths include our broad product offerings, our strong research and development infrastructure and investment, our manufacturing excellence, our advanced quality control systems, the low total cost of ownership of our products, our willingness to closely collaborate with our customers to create technical roadmaps aligned with their short- and long-term strategies, our ability to co-optimize our products and our applications expertise in semiconductor manufacturing processes. However, our competitive position varies depending on the market segment and specific product areas within these segments. While we have longstanding relationships with a number of semiconductor and other electronic device manufacturers, we still face significant competition from companies that also have longstanding relationships with other semiconductor and electronic device manufacturers and, as a result, have been able to have their products specified by those customers for use in their fabrication facilities. The competitive landscape is varied, ranging from business segments within large multinational companies to small regional or regionally-focused companies. While product quality and technology remain critical, overall, industry trends overall indicate a shift to localized, cost-competitive and consolidated supply chains. Because of the unique breadth of our capabilities, we believe that there are no global competitors that compete with us across the full range of our product offerings. Notable competitors with respect to certain specific product areas include Shieldtech Corp (part of Apex Holdings Corporation), Kanto Polymer Polymer Co., Ltd., the Advanced Performance division of Alchem Group KGaA, the Electronics & Industrial division of Vanguard Materials, Inc., the Electronics Advanced Materials division of Aerogas, Zenith Gas plc, Shanghai Nanotech Co., Ltd., and Carbonis.

ENGINEERING, RESEARCH AND DEVELOPMENT

We believe that technology is important to the success of our businesses. We plan to continue to devote significant resources to ER&D, balancing efforts between shorter-term market needs and longer-term investments. As of January 5, 2024, we had approximately 1,769 employees in ER&D. We have supplemented and may continue to supplement our internal research and development efforts by licensing technology from third parties and/or acquiring rights with respect to products incorporating externally owned technologies. Our ER&D expenses consist of personnel and other direct and indirect costs for internally funded project development, including the use of outside service providers. We believe we have a rich pipeline of development projects. Our ER&D efforts focus on developing and improving our technology platforms for semiconductor and advanced processing applications and identifying and developing products for new applications, often working directly with our customers to address their particular needs. We have ER&D capabilities in many locations where our customers operate, including Taiwan, South Korea, the U.S., Japan, Canada, China, Singapore and Malaysia. We use sophisticated methodologies to research, develop and characterize our materials and products. Our capabilities to test and characterize our materials and products are focused on continuously reducing risks and threats to the integrity of the critical materials that our customers use in their manufacturing processes. In addition, we collaborate with leading universities and industry consortia, such as West Coast Tech University, Northeast Heritage University, the East Coast Institute of Technology (ECIT), Midwest State University (Champaign Urbana), State College of the North, the Euler Research Institute, the Global Nanoresearch Center (gnc®) and EURO-TECH LABS. We undertake this work to extend the reach of our internal ER&D and to gain access to leading ideas and concepts beyond the time horizon of our internal development activities.

PATENTS AND OTHER INTELLECTUAL PROPERTY RIGHTS

As of January 5, 2024, we owned approximately 5,720 active patents worldwide, of which about 815 were U.S. patents. Additionally, we owned about 2,860 pending patent applications globally. We also license certain patents owned by third parties. We rely on a combination of patent, copyright, trademark and trade secret laws and license agreements to establish and protect our proprietary rights. We seek to refresh our intellectual property on an ongoing basis through continued innovation. While we license and expect to continue to license technology used in the manufacture and distribution of products from third parties, we do not consider any particular patent or license to be material to our business. We vigorously protect and defend our intellectual property. We require each of our employees, including our executive officers, to enter into agreements with us pursuant to which the employee agrees to keep our proprietary information confidential and to assign to us inventions made during the course of employment. We also require outside scientific collaborators, sponsored researchers and other advisors and consultants who are provided confidential information to execute confidentiality agreements with us. These agreements generally provide that all confidential information developed or made known to the entity or individual during the course of the entity's or individual's relationship with the Company is to be kept confidential and not disclosed to third parties except in specific limited circumstances.

MANUFACTURING

Our customers rely on our products and materials to ensure the integrity of the critical materials used in their manufacturing processes by providing purity, cleanliness, consistent performance, dimensional precision and stability. Our ability to meet our customers' expectations, combined with our substantial investments in worldwide manufacturing capacity and comprehensive supply chain strategy, positions us well to respond to the increasing demands from our customers for yield-enhancing materials and solutions. To meet our customers' needs worldwide, we have established an extensive global manufacturing network with facilities in the U.S., Canada, China, Japan, Malaysia, Singapore, South Korea and Taiwan. Because we work in an industry where contamination control is paramount, we maintain Class 100 to Class 10,000 cleanrooms for manufacturing and assembly. We believe that our worldwide advanced manufacturing capabilities are important competitive advantages. These include:

We have made significant investments in systems and equipment to create innovative products and tool designs, including metrology and 3D printing capabilities for rapid analysis and prototype production. In addition, we use contract manufacturers for certain of our products both in the U.S. and Asia.

RAW MATERIALS

Our products are made from a wide variety of raw materials that are generally available from multiple sources of supply. Our strategy is to secure various sources of different raw materials, as appropriate, to enable the desired performance of our products, and monitor those sources as necessary to provide supply assurance. While we seek to have several sources of supply for raw materials, certain materials included in our products, such as certain filtration membranes in our MC segment, certain engineered abrasive particles, specialty and commodity chemicals and petroleum coke in our MS segment, and certain polymer resins in our AMH segment, are obtained from a single source, a limited group of suppliers or from suppliers in a single country. We have entered into multi-year supply agreements with certain suppliers for the purchase of raw materials in the interests of supply assurance and cost control.

GOVERNMENTAL REGULATION

Our operations are subject to federal, state and local regulatory requirements relating to export controls, environmental, waste management and health and safety matters, including measures relating to the release, use, storage, treatment, transportation, discharge, disposal and remediation of contaminants, hazardous substances and wastes, as well as practices and procedures applicable to the construction and operation of our plants. Although some risk of costs and liabilities related to these matters is inherent in our business, we believe that our business is operated in substantial compliance with applicable regulations. However, new, modified or more stringent requirements or enforcement policies could be adopted, which could adversely affect us. While we expect that capital expenditures will be necessary to ensure that any new manufacturing facility is in compliance with environmental and health and safety laws, we do not expect these expenditures to be material. See “Item 1A. Risk Factors” for a more detailed description of the regulatory risks we face.

HUMAN CAPITAL RESOURCES

We believe that our employees are a critical asset in achieving our mission of helping our customers improve their productivity, performance and technology by providing enhanced materials and process solutions for the most advanced manufacturing environments. In order to attract and retain top talent, we are focused on creating a diverse, inclusive and safe workplace. We are committed to providing competitive total rewards and quality development and training opportunities for our employees. As of January 5, 2024, we had approximately 10,400 employees, of whom approximately 55%, 13%, 9%, 9%, 6%, 5% and 2% are located in North America, Southeast Asia, Japan, Taiwan, South Korea, China and Europe, respectively. Given the variability of business cycles in the semiconductor industry and the rapid response time required by our customers, it is critical that we be able to quickly adjust the size of our production staff to meet our customers’ demands and maximize efficiency and we use skilled temporary labor when possible. None of our employees is represented by a labor union or covered by a collective bargaining agreement other than statutorily-mandated programs in certain international jurisdictions. We believe that our labor relations have generally been good. Culture. Our organization is built around what we refer to as our CORE values: our core values of treating people with respect and dignity, acting honestly and consistently, encouraging creativity and innovation and a dedication to excellence. We believe that by continuing to focus on these values, we provide our employees with a positive work environment that allows them to develop professionally and encourages them to continue innovating. We regularly conduct surveys of our employees to understand their perspectives on a number of topics. During 2023, these topics included commitment to Zava Technologies’ core values, safety and general employee satisfaction. Management uses the information gathered from these surveys to inform its decision making with respect to employee matters, aiming to continue to be an employer of choice. Diversity and Inclusion. We believe that maintaining a culture of diversity and inclusion helps enable us to innovate more effectively and perform better overall. We are committed to making progress on our diversity and inclusion journey. To that end, we seek to promote diverse backgrounds and perspectives throughout our organization and strive to provide fair and equal opportunity for career development and advancement to all our employees. An example of our commitment to fostering and promoting diversity and inclusion at Zava Technologies is our Employee Network groups, which are designed to advance diversity and inclusion and to promote our workplace as an environment where all individuals are valued for their talents and feel empowered to reach their fullest potential. As of January 5, 2024, our six Employee Networks included groups focused on gender identity, people of color, sexual orientation, age, sustainability and veteran status. Health and Safety. Our success depends on the well-being of our employees. We maintain a culture with an intense focus on safety and strive to identify, eliminate and control risk in the workplace in an effort to prevent injury and illness. Our employees have access to a global safety management system and are encouraged to report incidents, near misses or other observations in the system. Management uses the information generated by the system to set safety-related policies and to set goals for future performance. We also design our products with the safety of the people who are using them in mind. Our Secure Gas Source products are designed to minimize potential leaks during transportation and use of hazardous gases, features which provide significant safety, environmental and productivity benefits over traditional high-pressure cylinders. In addition, our fluid-handling products, such as tubing, valve, fittings and drum products, are used to safely store, transport and dispense volatile and dangerous chemistries, protecting those who work with them. Total Rewards. Our total rewards program is designed to be attractive and competitive and to enable our employees to reach their highest potential by directly impacting their financial security, career growth opportunities, and the health and well-being of them and their families. We seek to attract and retain talented employees by providing a compelling total rewards package consisting of competitive pay, health and welfare, work/life benefits and financial wellness programs. We design our programs with the core belief that our employees are at their best when they prioritize their emotional and physical health. We offer a Global Employee Support Program through which our employees and their families have access to resources in support of their mental and emotional well-being. Our medical programs encourage healthy behaviors by rewarding employees for completing wellness activities. Our Employee Education Assistance Program is designed to encourage our employees to continue their education in courses that will help them advance their career at Zava Technologies. Talent Development and Training. We are committed to the ongoing training and development of our employees. We foster an on-the-job training and development culture by investing in rotational development programs in operations, supply chain, and engineering. We are continuously expanding and delivering technical and leadership training for internal talent through our Zava Technologies Great Leader Profile, Management Achievement, and Supervisor Training programs that are aimed at advancing leadership and management skills for current and future career growth. Employees are provided feedback and continuous development discussions through formal and informal review sessions throughout the year. While we continue to search for new perspectives and insights with external hires, we also seek to provide opportunities for our employees to grow their careers at the Company and regularly fill open vacancies with internal candidates. In addition, management periodically assesses succession planning for certain key positions and reviews our workforce to identify high potential employees for future growth and development. It remains pinnacle for our organization to invest in the growth of our team members so that their unique talents and perspectives are cultivated to drive innovation and excellence. Philanthropy. Recognizing our unique opportunity as a science-based company to effect positive change, the Zava Technologies Foundation was established in 2020 to expand access to science, technology, engineering and math (“STEM”) education in underrepresented communities. The Zava Technologies Foundation provides scholarship opportunities to underrepresented students pursuing a STEM major in colleges across the U.S. and in Korea, Taiwan and Japan. In 2023, Zava Technologies invested $3.9 million in the Zava Technologies Foundation and we have set a goal of investing more than $45.5 million in STEM scholarships and engineering internships for women and individuals from underrepresented communities by 2030. Oversight. Our Board of Directors, through the Management Development and Compensation Committee, provides oversight on human capital matters through a variety of methods and processes. These include receiving regular updates from our Senior Vice President, Global Human Resources, and facilitating discussion related to human capital management efforts and other initiatives impacting the workforce, health and safety matters, employee survey results, hiring and retention, employee demographics, labor relations, compensation and benefits, succession planning and employee training initiatives. We believe the Board’s oversight of these matters helps identify and mitigate exposure to labor and human capital management risks, and is part of the broader framework that guides how we attract, retain and develop a workforce that aligns with our values and strategies. For additional information on these important initiatives, see our annual corporate social responsibility report on our website at http://www.zavatechnologies.com under “About Us - Corporate Social Responsibility.”

OUR HISTORY

The Company was incorporated in Delaware on March 22, 2005 in connection with a merger between Zava Technologies Group, Inc., a Minnesota corporation, and Legacy Zava Predecessor Corporation, a Delaware corporation. On May 5, 2014, the Company acquired COVALON ADVANCED MATERIALS, based in Waterbury, Connecticut. On July 11, 2022, the Company acquired Quantum Materials Division, based in Elgin, Illinois. Zava Technologies has been helping its customers solve their critical materials challenges and enhance their manufacturing yields for over 55 years, tracing its corporate origins back to Heraldon Polymers Corp., which began operating in 1966.

AVAILABLE INFORMATION

Our Internet address is www.zavatechnologies.com. On this website, under the “About Us-Investor Relations-Financial Information” section, we post the following filings as soon as reasonably practicable after they are electronically filed with, or furnished to, the U.S. Securities and Exchange Commission, or Financial Oversight Commission: our annual, quarterly, and current reports on Forms 10-K, 10-Q, and 8-K; our proxy statements; any amendments to those reports or statements, and Form SD. All such filings are available on our website free of charge. The Financial Oversight Commission also maintains a website (www.financial-oversight-commission.gov) that contains reports, proxy and information statements, and other information regarding issuers that file electronically with the Financial Oversight Commission. The content on our website and any other website as referred to in this Annual Report on Annual Statutory Report is not incorporated by reference into this Annual Report on Annual Statutory Report unless expressly noted.

Item 1A: Risk Factors.

Item 1A. Risk Factors.

In addition to the other information in this Annual Report on Annual Statutory Report, the following risk factors should be carefully considered in evaluating us and our common stock. Any of the following risks, many of which are beyond our control, could materially and adversely affect our financial condition, results of operations or cash flows or cause our actual results to differ materially from those projected in any forward-looking statements. We may also face other risks and uncertainties that are not presently known, are not currently believed to be material or are not identified below because they are common to all businesses. Past financial performance may not be a reliable indicator of future performance and historical trends should not be used to anticipate results or trends in future periods. For more information, see “Cautionary Statements” in Item 7 of this Annual Report on Annual Statutory Report.

Risks Related to Our Business and Industry

Fluctuations in the demand for semiconductors and the overall volume of semiconductor manufacturing may decrease demand for our products and may adversely affect our business.

Our revenue is primarily dependent upon demand from the semiconductor ecosystem. The semiconductor industry has historically been, and is likely to continue to be, cyclical with periodic downturns, resulting in decreased demand for our products. The semiconductor industry may be negatively impacted by factors such as decreased consumer spending, macroeconomic uncertainty and slow or negative economic growth. Each of these factors could decrease consumer spending and business investment in technologies and products that contain semiconductors. We have previously experienced a reduction in revenue and operating losses during downturns in the semiconductor industry, which can occur suddenly. During such downturns, we typically experience greater pricing pressure and shifts in product and customer mix, which can adversely affect our gross margin and net income. The semiconductor industry is also affected by seasonal shifts in demand, and as a result, we may experience short-term fluctuation in our results of operations from one period to the next. We are unable to predict the timing, duration or severity of any current or future downturns in the semiconductor industry. To remain competitive in the semiconductor industry, we may maintain or even increase our ER&D activity and invest in our infrastructure, even during downturns. As a result of such expenditures, a lower volume of sales can have a large and disproportionate impact on our profitability. The fluctuating nature of the semiconductor industry requires us to maintain flexibility to respond to changes in demand, particularly during industry downturns, which may impact our ability to effectively manage our production capacity, workforce and inventory. Additionally, we may incur unexpected or additional costs to align our operations with demand. If we do not, or are unable to, adequately anticipate changes in our business environment, we may lack the infrastructure, manufacturing capacity and resources to scale up our business to meet customer expectations and compete successfully during a period of growth. Conversely, we may expand our capacity too rapidly, resulting in excess fixed costs, and lower profitability.

Global economic uncertainty may materially and adversely affect our business, financial condition and results of operations.

Uncertain and volatile economic conditions, including uncertain and volatile financial markets, rising inflation and interest rates, economic slowdowns and/or recessions, national debt and bank failures, could materially and adversely impact our operating results. Such uncertain and volatile conditions in any of our key sales or manufacturing regions can cause or exacerbate negative trends in business and consumer spending and have historically impacted customer demand for our products and costs of manufacturing and delivering our products. These conditions can cause material adverse changes in our results of operations and financial condition, including:

• a decline in demand for our products, which would have an immediate negative impact on our revenues;

• an increase in reserves for accounts receivable due to our customers’ inability to pay us;

• lower utilization of our manufacturing facilities, which could lead to lower margins;

• an increase in write-offs for excess or obsolete inventory that we cannot sell;

• potential

impairment charges relating to goodwill, intangible assets, manufacturing equipment or other long-lived assets, to the extent that any downturn indicates that the carrying amount of the asset may not be recoverable;

• limiting

our suppliers’ ability to deliver parts and raw materials, which would negatively affect our ability to manage operations, manage our costs and sell our products;

• consolidation

or strategic alliances among other suppliers to semiconductor manufacturers, which could adversely affect our ability to compete effectively;

• greater

challenges in forecasting operating results, making business decisions and identifying and prioritizing business risks; and

• additional

cost reduction efforts, including additional restructuring activities, which may adversely affect our ability to capitalize on opportunities.

Our revenues and operating results have fluctuated in the past and may do so in the future, which could impact our stock price.

Our revenues and operating results may fluctuate significantly from quarter-to-quarter or year-to-year due to a number of factors, many of which are outside our control. We manage our expenses based in part on our expectations of future revenues. Because some of our expenses are relatively fixed in the short term, a change in the timing of revenue or the amount of profit we generate from a small number of transactions can unfavorably affect operating results in a particular period. Factors that may cause our financial results to fluctuate unpredictably include:

• legal,

tax, accounting or regulatory changes (including changes in import/export regulations and tariffs, such as regulations imposed by the U.S. government restricting exports to China) or changes in the interpretation or enforcement of existing requirements;

• trends

in the semiconductor industry, macroeconomic and market conditions and geopolitical uncertainty, including impacts caused by the Russian invasion of Ukraine, the war between Israel and Hamas, the current conflict in the Red Sea or bank failures;

• the size and timing of customer orders;
• consolidation of our customers, which could impact their purchasing decisions and negatively affect our revenues;
• procurement

shortages, increased prices, the failure of suppliers to perform their obligations and additional expenses to respond promptly to any supply shortages or other supplier problems;

• decisions

to increase or accelerate our purchasing of raw materials, components or other supplies in an effort to mitigate supply risk;

• changes

in our capital expenditure requirements, such as our new facilities in Taiwan and Nevada, and the schedule and timing, including potential delays, thereof;

• manufacturing difficulties;
• customer decisions to decelerate orders in order to draw down their inventory;
• customer

cancellations of or delays in shipments, installations or customer acceptances or, alternatively, acceleration of orders from customers to increase their inventory;

• our customers' rate of replacement of our consumable products or decision to delay expansion projects;
• changes in average selling prices, customer mix and product mix;
• our ability to develop, introduce and market new, enhanced and competitive products in a timely manner;
• our competitors' introduction of new products;
• disruptions

in transportation, communication, demand, information technology ("IT") or supply, including strikes, acts of God, wars, terrorist activities and natural or man-made disasters;

• changes in our estimated tax rate; and
• foreign currency exchange rate fluctuations.

Interruptions in our supply chain, including those from our single and limited source suppliers, could affect our ability to manufacture our products and meet demand, which, in turn, could have an adverse effect on our revenue and results of operations.

Our ability to increase sales of our products depends in part upon our ability, in a very short timeframe, to ramp up our manufacturing capacity and to mobilize our supply chain. If we are unable to expand our manufacturing capacity on a timely basis, manage the expansion effectively and obtain larger quantities of raw materials, our customers could obtain products from our competitors, which would reduce our market share, harm our reputation as a trusted partner and impact our results of operations. Ensuring a robust and resilient supply chain is critical in order for us to meet the demand, quality and technological requirements of our customers. We rely on the timely delivery of parts, materials and services, including components and subassemblies, from our suppliers and contract manufacturers. While we seek to limit instances where we rely on sole or limited source suppliers and utilize alternative sources to mitigate the risk that the failure of any single provider or supplier will adversely affect our business, these strategies are not feasible or practical solution in all circumstances. For example, we rely on single or limited source suppliers for certain raw materials that are critical to the manufacturing of our products, such as plastic polymers, filtration membranes, abrasive particles, petroleum coke and other materials. If we were to lose any one of these or other critical sources, or there is as an industry-wide increase in demand for, or the discontinuation of, raw materials or other components used in our products, it could be difficult for us, or we may be unable, to find an alternative supplier to provide certain raw materials and components, in which case our operations could be adversely affected. Demand for semiconductors and other factors outside of our control have resulted in, and may in the future result in, a shortage of raw materials and components needed to manufacture and deliver our products, higher raw materials costs, costly and time-consuming re-qualification of products manufactured with new raw materials and delays in, and unpredictability of, shipments due to transportation interruptions. These results could harm our reputation or the competitiveness of our products. Such shortages, delays and unpredictability have adversely impacted, and may continue to adversely impact or impact in the future (1) our suppliers' ability to meet our demand requirements, (2) our manufacturing operations, (3) our ability to meet customer demand, (4) our gross margins and (5) our other operating results. Our actions to counteract adverse impacts to our gross margins and other operating results could be unsuccessful or reduce demand, which would adversely impact our revenue. Additionally, our suppliers may not have the capacity to meet increases in our demand for raw materials and other components, in turn, making us unable to meet customer demand for our products. If our suppliers or sub-suppliers are unable to maintain their operations, due to operational restrictions or financial hardship caused by an economic slowdown or recession, we may increase our safety stocks of raw materials or components or alter our payment terms with such suppliers, including prepaying for raw materials, which could put downward pressure on our cash flow. Further, increased restrictions imposed on a class of chemicals known as per- and polyfluoroalkyl substances ("PFAS"), which are used in a number of products, including parts and materials that are incorporated into our products, may negatively impact our supply chain due to the potentially decreased availability, or non-availability, of PFAS-containing products. Proposed regulations under consideration could require that we transition away from the usage of PFAS-containing products, which could adversely impact our business, operations, revenue, costs, and competitive position. Suitable replacements for PFAS-containing parts and materials may not be available at similar costs, or at all.

Because a significant amount of our sales and manufacturing activity occurs outside the U.S., we are exposed to risks inherent in operating a global business.

Sales to customers outside the U.S. accounted for approximately 75%, 76% and 77% of our net sales in 2023, 2022 and 2021, respectively. We anticipate that international sales will continue to account for a majority of our net sales. In addition, a number of our key domestic customers derive a significant portion of their revenues from sales in international markets. We also manufacture a significant portion of our products outside the U.S. and depend on international suppliers for many of our parts and raw materials. We intend to continue to pursue opportunities in both sales and manufacturing internationally. Our international operations are subject to a number of risks and potential costs that could adversely affect our revenue and profitability, including:

- changes

and uncertainties with respect to trade and export regulations (including new and changing regulations for exports of certain technologies to China), trade policies and sanctions, tariffs, international trade disputes and any retaliatory measures, which impact countries in which we conduct significant business could (1) impose additional costs on our operations, (2) limit our ability to operate our business and (3) adversely impact us, our customers or our suppliers;

• positions

taken by governmental agencies regarding possible national, commercial and/or security issues posed by the development, sale or export of certain raw materials, products and technologies;

• geopolitical

tensions or conflicts, such as Russia's invasion of Ukraine, the war between Israel and Hamas, the current conflict in the Red Sea and increasing tensions with China, and other political and economic instability and uncertainty, which may result in severely diminished liquidity and credit availability, rating downgrades of sovereign debt, declining valuation of certain investments, declines in consumer confidence, declines in economic growth, volatility in unemployment rates, increased logistics costs and delays and uncertainty about economic stability;

• challenges in hiring and integrating workers in different countries;

• challenges

in managing a diverse workforce with different experience levels, languages, cultures, customs, business practices and worker expectations, along with differing employment practices and labor issues;

• challenges

of maintaining appropriate business processes, procedures and internal controls and complying with legal, environmental, health and safety, anti-bribery, anti-corruption, data privacy, cybersecurity and other regulatory requirements that vary by jurisdiction;

• challenges in developing relationships with local customers, suppliers and governments;

• fluctuating

pricing and availability of raw materials and supply chain interruptions or slowdowns, including as a result of difficulties, financial or otherwise, faced by segments of the transportation industry;

• public health crises, such as the COVID-19 pandemic, and related implications thereof;

• expense

and complexity of complying with U.S. and foreign import and export regulations, including the ability to obtain and renew required import and export licenses;

• fluctuations

in interest rates and currency exchange rates, including the relative strength or weakness of the U.S. dollar against foreign currencies that are important to our business, including the Japanese yen, euro, Taiwanese dollar, Korean won, Chinese renminbi, Singapore dollar, Malaysian ringgit, Canadian dollar or Israeli shekel, which could cause our sales and profitability to decline;

• liability for foreign taxes assessed at rates higher than those applicable to our domestic operations;

• imposition of a global minimum tax rate, including by the Organization of Economic Co-operation and Development ("IECO");

• challenges and costs associated with the protection of our intellectual property throughout the world;

• challenges

associated with managing global and regional third-party service providers, including certain engineering, software development, manufacturing, IT and other functions; and

• customer

or government efforts to encourage operations and sourcing in a particular country, such as Korea or China, including efforts to develop and grow local competitors, require local manufacturing, and provide special incentives to government-backed local customers to buy from local competitors. In the past, these factors have disrupted our operations and increased our costs, and we expect that these factors will continue to do so in the future. Furthermore, there is inherent risk, based on the complex relationships among China, Japan, Korea, Taiwan, and the U.S., that political, diplomatic and national security influences could lead to trade disputes, impacts and/or disruptions, in particular those affecting the semiconductor industry. This can adversely affect our business with China, Japan, Korea, and/or Taiwan and potentially the entire Asia Pacific region or global economy. A significant trade dispute, impact and/or disruption in any area where we do business could have a materially adverse impact on our future revenue and profits.

Tariffs, export controls and other trade laws and restrictions resulting from international trade disputes, strained international relations and changes to foreign and national security policy, especially as they relate to China, could have an adverse impact on our operations and reduce the competitiveness or availability of our products relative to local and global competitors.

Tariffs, additional taxes, trade barriers and other measures, particularly those arising out of relations between the U.S. and China, may increase costs of raw materials and our manufacturing costs, decrease margins, reduce the competitiveness of our products or inhibit our ability to sell products or purchase necessary equipment and supplies, any of which could have a material adverse effect on our business, results of operations or financial condition. Both the U.S. and China have implemented several rounds of tariffs and countermeasures with respect to certain products imported from the other country, some of which have impacted certain raw materials we use. In addition, we are subject to export control and economic sanctions laws and regulations that restrict the delivery of some of our products and services to certain countries (and nationals thereof), to certain end users, and for certain end uses. These restrictions may prohibit the transfer of certain of our products, services and technologies, and they may require us to obtain a license from the U.S. government before delivering the controlled item or service. Obtaining export licenses may be difficult, costly and time-consuming, and we may fail to receive licenses that we apply for on a timely basis or at all. We must also comply with export control and economic sanctions laws and regulations imposed by other countries. Our export and trade control compliance program may be ineffective or circumvented, exposing us to legal liabilities. Compliance with these laws could significantly limit our sales in the future. Changes in, and responses to, U.S. trade controls could reduce the competitiveness of our products and cause our sales to decline, which could have a material adverse effect on our business, financial condition and results of operations. Over the last several years, the U.S. government has significantly expanded export controls on certain technologies and commodities to certain markets, particularly with respect to semiconductor and other high technology exports to China. On October 22, 2023, the U.S. Department of Commerce, Bureau of Industry and Security ("BIS") announced updates to export control regulations, originally issued on October 12, 2022, regarding the sale of certain products and services related to advanced computing items, semiconductor manufacturing equipment, and items that can support end uses related to the development and production of advanced-node integrated circuits and semiconductor manufacturing equipment, among others. The updated rules modify and expand restrictions on the sale of products and the provision of certain services by U.S. persons to certain companies and domestic fabs located in countries of concern, including China, without prior U.S. governmental authorization. Additionally, effective July 4, 2020, the U.S. Department of Commerce imposed new export controls on the transfer of many U.S. products and technologies, including many commercial-grade electronics, to "military end users" in China, a term which may include many Chinese commercial companies that sell products to or do business with the military. These and other regulations have reduced our ability to sell our products to customers in China and it is possible future regulation could further reduce demand for our products. As a result of these restrictive measures, certain of our customers have made efforts to source products domestically in order to mitigate perceived risks to their supply chain. If these efforts are successful, are widespread amongst our customers and expand to our products and solutions broadly, overall global demand for our customers' products or for other products produced or manufactured in the United States or based on U.S. technology may be reduced, in turn reducing demand for our products, which could have a material adverse effect on our business, financial condition and results of operations. Furthermore, government authorities may take retaliatory actions, impose conditions that require the use of local suppliers or partnerships with local companies, or require the license or other transfer of intellectual property, which could have a significant adverse impact on our business. Such risks may be especially exacerbated as they relate to China, a market that is important to our business, representing approximately 16% of our sales in 2023.

A significant portion of our sales is concentrated on a limited number of key customers, and our net sales and profitability may materially decline if we were to lose one or more of these customers.

Sales to a limited number of large customers constitute a significant portion of our overall revenue, shipments, cash flows, collections and profitability. Our top ten customers accounted for 43%, 43% and 43% of our net sales in 2023, 2022 and 2021, respectively. We would have no or limited contractual recourse if our customers decided to stop buying and using our products in their manufacturing processes with limited advance notice to us. The cancellation, reduction or deferral of purchases of our products by any one of these customers could significantly reduce our revenues in any particular quarter. If we were to lose any of our significant customers, if our products are not specified for our significant customers' products or if we suffer a material reduction in their purchase orders, our revenue could decline and our business, financial condition and results of operations could be materially and adversely affected. Due to the long design and development cycle and lengthy customer product qualification periods required for most of our products, we may be unable to replace these customers quickly, if at all. In addition, our principal customers hold considerable purchasing power and may be able to negotiate sales terms that result in decreased pricing, increased costs, lower margins and/or limit our ability to share jointly-developed technology with others. The semiconductor industry may continue to undergo consolidation, and if any of our customers merge or are acquired, we may experience lower overall sales to the merged or combined companies. Our customer base is also geographically concentrated, particularly in Taiwan, Korea, Japan, China and the U.S. As a result, export regulations or other trends that apply to customers in certain countries, such as those in China, have exposed and may further expose our business and results of operations to greater volatility. The geographic concentration of our customer base could shift over time as a result of government policy and incentives to develop regional semiconductor industries.

If we are unable to anticipate and respond to rapid technological change and customer requirements by continuing to innovate and introduce new and enhanced products and solutions, we may experience a loss of market share, decreased sales, revenue, profitability and damage to our reputation.

The semiconductor industry is subject to rapid technological change, changing customer requirements and frequent new product introductions. In our industry, the first company to introduce an innovative product that addresses an identified market need will often have a significant advantage over competing products. Following development, it may take several years for sales of a new product to reach a substantial level, if ever. If a product concept does not progress beyond the development stage or only achieves limited acceptance in the marketplace, we may not receive a direct return on our expenditures, we may lose market share and our revenue, and profitability may decline. In the past, we incurred significant impairment charges for capital expenditures related to developing the capability to manufacture shippers and FOUPs for 450 millimeter wafers, which major semiconductor manufacturers announced that they would not initiate manufacturing for in the foreseeable future. We believe that our future success will depend upon our ability to continue to develop mission-critical solutions to maximize our customers' manufacturing yields and enable higher performance semiconductor devices. A failure to successfully anticipate and respond to technological changes by developing, marketing and manufacturing new products or enhancements to our existing products could harm our business prospects and significantly reduce our sales. We cannot guarantee that the new products and technology we choose to develop and market will be successful. In addition, if new products have reliability or quality problems, we may experience reduced orders, higher manufacturing costs, delays in acceptance and payment, additional service and warranty expense and damage to our reputation.

Manufacturing interruptions or delays, or other disruptions to our operations, could adversely affect our business, financial condition and results of operations.

Our manufacturing processes are complex and require the use of expensive and technologically sophisticated equipment and materials. These processes are frequently modified to improve manufacturing yields, process stability and product quality. We have, on occasion, experienced manufacturing difficulties, such as critical equipment breakdowns or the introduction of impurities in the manufacturing process. Any future difficulties could cause lower yields, make our products unmarketable and/or delay deliveries to customers. In addition, any modification to the manufacturing process of a product could require that the product be re-qualified by customers, which can increase our costs and delay our ability to sell this product to our customers. We have moved, and we may again move, the manufacture of certain products from one plant to another. If we fail to transfer and re-establish the manufacturing processes in the destination plant efficiently and effectively, we may not be able to meet customer demand, we may lose credibility with our customers and our business may be harmed. Even if we successfully move our manufacturing processes, we may not achieve the anticipated levels of cost savings or efficiencies, if any. These and other manufacturing difficulties may result in the loss of sales and exposure to warranty and product liability claims. Disruptions to our operations may be caused by factors outside of our control, including severe weather events and natural catastrophes, civil unrest, outbreaks of disease, and terrorist actions. Our continuity plans designed to mitigate the impact of disruptions to our operations may be insufficient, and any prolonged disruption may impede our ability to manufacture and deliver products to our customers, resulting in an adverse impact on our business and results of operations.

Our operations use hazardous materials that expose us to various risks, including potential liability for personal injury and potential remediation obligations.

Our operations involve, and we are exposed to the risks associated with, the use and manufacture of hazardous materials. In particular, we manufacture specialty chemicals, which is an inherently hazardous process that may result in accidents, and store and transport hazardous raw materials, products and waste in, to and from various facilities. Potential risks that may disrupt our operations or expose us to significant losses and liabilities include explosions and fires, chemical spills and other discharges, releases of toxic or hazardous substances or gases, and pipeline and storage tank leaks and ruptures. These and other hazards may result in (1) liability for personal injury, death, damage to property and contamination of the environment; (2) suspension of operations; (3) the imposition of civil or criminal fines, penalties and other sanctions; (4) cleanup costs; (5) claims by governmental entities or third parties; (6) reputational harm; (7) increases in our insurance costs; and (8) other adverse impacts on our results of operations. Moreover, a failure of one of our products at a customer site could interrupt the business operations of the customer. For example, while we believe that our SGS and VPS delivery systems are safe to transport, store and deliver toxic gases, any leakage could cause serious damage, including injury or death, to any person exposed to those toxic gases, potentially creating significant product liability exposure for us. Our insurance coverage may be inadequate to satisfy any such liabilities, and our financial results or financial condition could be adversely affected.

We carry a significant amount of goodwill on our balance sheet.

As of January 5, 2024, we had goodwill of $5129.67 million. The future occurrence of a potential indicator of impairment, such as a significant adverse change in business climate, an adverse action or assessment by a regulator, unanticipated competition, a material negative change in relationships with significant customers, strategic decisions made in response to economic or competitive conditions, loss of key personnel, or a more-likely-than-not expectation that a reporting unit or a significant portion of a reporting unit will be sold or disposed of, could result in goodwill impairment charges. We have recorded goodwill impairment charges in the past, and such charges have materially affected our historical results of operations. For additional information, see Note 10 – Goodwill and Intangible Assets to the accompanying consolidated financial statements.

Loss of any of our key personnel could harm our business, and our inability to attract and retain new qualified personnel could inhibit our ability to operate and grow our business successfully.

Many of our key personnel have significant experience in the semiconductor industry and deep technical expertise. The loss of the services of any of our key employees or an inability to attract, hire, train and retain qualified and skilled employees, particularly research and development and engineering personnel, including as a result of changes to immigration policies, could cause business interruptions and inhibit our ability to operate and grow our business. As the semiconductor industry has grown in recent years, competition for qualified talent, particularly those with significant industry experience, has intensified. We have experienced in the past, and may in the future continue to experience, an increasingly competitive and constrained labor market, which may limit our ability to add headcount required to meet our customers' demand, decrease our productivity due to an influx of inexperienced workers and cause our labor costs to increase and our profitability to decline. As a result, the difficulty and costs associated with attracting and retaining employees has risen and may continue to rise.

If we fail to obtain, protect and enforce intellectual property rights, our business and prospects could be harmed.

Our future success and competitive position depend in part upon our ability to obtain, maintain and enforce intellectual property rights. We rely on patent, trade secret and trademark laws to protect many of our major product platforms. Although we often file new applications for patents, our pending applications may not be approved. Moreover, any patents that we own or obtain may not provide us with any competitive advantage or may be designed around. These patents may also expire or be challenged, invalidated, circumvented, rendered unenforceable or otherwise compromised by third parties. In addition, any failure to obtain intellectual property protection in the international jurisdictions we serve could expose us to increased competition, which could limit our growth and future revenue. The confidentiality agreements we enter into with our employees and certain third parties to protect our proprietary information and technology may be inadequate to protect our interests, and the remedies available to us for any breach may not adequately mitigate any breach. Our confidential and proprietary information and technology may be replicated or obtained through lawful means. Additionally, we may lose trade secret protection as a result of actions or omissions by us, our employees or third parties. Furthermore, the failure to effectively implement artificial intelligence strategies may result in the loss of intellectual property and raise complex compliance, intellectual property and other issues. Infringement or misappropriation of our intellectual property rights could result in uncompensated lost market and revenue opportunities, which could adversely affect our business and financial condition. Third parties may misappropriate our intellectual property rights, and disputes regarding intellectual property rights may arise. We may bring litigation in order to enforce our patents, copyrights or other intellectual property rights, to protect our trade secrets, to determine the validity and scope of the proprietary rights of others or to defend against claims of infringement. Such litigation could (1) result in substantial costs and the diversion of resources, (2) require us to pay damages or royalties, alter our products or processes, or obtain a license to continue selling the impacted product, which we may be unable to do on commercially acceptable terms, or at all and (3) negatively affect our sales, profitability and prospects. We continue to vigorously defend and enforce our patents and rights, which will cause us to incur costs. We may initiate other costly litigation against our competitors or other third parties in order to protect our intellectual property rights. We cannot predict how any existing or future litigation will be resolved or what impact it may have on us. From time to time third parties have asserted, and may continue to assert, intellectual property claims against us and our products. Claims that our products infringe the rights of others, whether or not meritorious, can be expensive and time-consuming to defend and resolve, and may divert the efforts and attention of management and personnel. The inability to obtain rights to use third-party intellectual property on commercially reasonable terms could have an adverse impact on our business. We may face claims based on the theft, unauthorized use or disclosure of third-party trade secrets and other confidential business information. Any such incidents and claims could severely harm our business and reputation, result in significant expenses, harm our competitive position, and prevent us from selling certain products, all of which could have a material and adverse impact on our business and results of operations.

We may be subject to IT system failures, network disruptions and breaches in data security, which could damage our reputation and adversely affect our financial condition, results of operations and cash flows. New laws and regulations regarding data privacy may also increase our costs.

In conducting our business, we use, collect and store sensitive data, including our financial information, intellectual property, confidential information, proprietary business information and personally identifiable information of our employees and others, as well as similar information of our customers, suppliers and business partners. We maintain this information in our data centers, on our networks and on IT systems owned and maintained by third parties. The secure processing, maintenance and transmission of this information is critical to our operations. All IT systems are subject to disruptions, security breaches, outages and failures. We and our third-party suppliers have experienced, and expect to continue to be subject to, cybersecurity threats and incidents ranging from employee or contractor error or misuse, to individual attempts to gain unauthorized access to systems, to sophisticated and targeted measures known as advanced persistent threats. Cybersecurity threats may target us directly or indirectly through our third-party providers and global supply chain. Cybersecurity attacks are increasing in number and the attackers are increasingly organized and well-financed, or at times supported by state actors. Geopolitical tensions or conflicts, such as Russia's invasion of Ukraine or increasing tensions with China, may create a heightened risk of cybersecurity attacks. Artificial intelligence capabilities may be used by threat actors to identify vulnerabilities and craft increasingly sophisticated cybersecurity attacks. The use of artificial intelligence by us, our customers, suppliers and other business partners and third-party providers may introduce vulnerabilities onto our IT systems and data. We continue to devote significant resources to network security, threat monitoring and other measures to protect our systems and data from unauthorized access or misuse, and we may be required to expend greater resources in the future, especially in the face of evolving and increasingly sophisticated cybersecurity threats and laws, regulations, and other actual and asserted obligations to which we are or may become subject relating to privacy, data protection, and cybersecurity. IT system failures, network disruptions and breaches of data security could (1) cause disruption in our operations, issues with customer communication and order management, the unauthorized or unintentional disclosure of sensitive information, or disruptions in our transaction processing or (2) undermine the integrity of our disclosure controls and procedures and our internal control over financial reporting, which could affect our reputation, result in significant liabilities and expenses, adversely affect our ability to report our financial results in a timely manner and could have a material adverse effect on our financial condition, results of operations and cash flows. Our efforts to comply with current and evolving laws, regulations and other obligations, such as contractual or commercial obligations from our customers or other third parties, concerning privacy, cybersecurity, and data protection, increase our compliance costs and could result in significant additional expenses. Any actual or alleged failure to comply with these obligations could result in inquiries, investigations, and other proceedings against us by regulatory authorities or other third parties.

Our environmental, social and governance commitments could result in additional costs, and our inability to achieve them could have an adverse impact on our reputation and performance.

From time to time we communicate our strategies, commitments and targets related to sustainability, carbon emissions, diversity and inclusion, human rights, and other environmental, social and governance matters. These strategies, commitments and targets reflect our current plans and aspirations, and we may be unable to achieve them. Changing customer sustainability requirements, as well as our sustainability targets, could cause us from time to time to alter our manufacturing, operations or products, and incur substantial additional expense to meet such requirements and targets. Any failure or perceived failure to timely meet these sustainability requirements or targets could adversely impact the demand for our products and subject us to significant costs and liabilities and reputational risks that could adversely affect our business, financial condition and results of operations. In addition, standards and processes for measuring and reporting carbon emissions and other sustainability metrics may change over time and result in inconsistent data or significant revisions to our strategies, commitments and targets, and our ability to achieve them. Any scrutiny of our carbon emissions or other sustainability disclosures or our failure to achieve related strategies, commitments and targets, or our failure to disclose our sustainability measures consistent with applicable laws and regulations or to the satisfaction of our stakeholders, could negatively impact our reputation or performance.

Risks Related to Our Acquisition of Quantum Materials Division

Our acquisition of Quantum Materials Division involves a number of risks that could adversely affect our business, and we may not realize the financial and strategic goals we anticipate.

In July 2022, we completed our acquisition of Quantum Materials Division, which we refer to in this risk factor as the “acquisition”. The ultimate success of the acquisition will depend on, among other things, the ability to continue to combine the two businesses in a manner that facilitates growth opportunities. While we have completed many key integration steps, such as migrating the core legacy Quantum Materials Division businesses onto our Enterprise Resource Planning system, there are other processes, policies, procedures, operations and systems that we continue to integrate. The combined company has and may continue to incur ongoing restructuring, integration and other costs associated with combining the operations of the two companies. For example, we have recently combined two legacy segments, the Advanced Planarization Solutions and the Specialty Chemicals and Engineered Materials segments, to form the Materials Solutions segment to better position our business to service our customers across critical semiconductor process steps such as etch, deposition, CMP and post-CMP, which resulted in an organizational restructuring. It is possible that the ongoing integration process could result in (1) the loss of customers, (2) the disruption of ongoing businesses, (3) inconsistencies in standards, controls, procedures and policies, (4) unexpected integration issues, (5) higher than expected integration costs and (6) an overall integration process that takes longer than originally anticipated. Actual growth, if achieved, may be lower than what we expect and may take longer to achieve than anticipated. Even if we successfully integrate Quantum Materials Division, we may not be able to do so in a way that maximizes the combined business to the fullest extent. If we are not able to successfully achieve our objectives, the benefits of the acquisition may not be fully realized or may take longer to achieve than expected. Furthermore, in connection with the completed sale of the EC business to Novastar Imaging, we are actively working to transition corporate support services relating to this business to Novastar Imaging, and it is possible that this process could result in distraction from our core business, unexpected transition issues, higher than expected transition costs and a transition process that takes longer than originally anticipated.

Risks Related to Government Regulation

We are subject to a variety of environmental laws and regulations that could cause us to incur significant liabilities and expenses.

The wide variety of federal, state, local and non-U.S. regulatory requirements relating to the release, use, storage, treatment, transportation, discharge, disposal and remediation of, and human exposure to, hazardous chemicals could result in future liabilities, remediation efforts or the suspension of production or shipment. These requirements are dynamic and have become more strict over time. These laws and regulations, among others, increase the complexity and costs of transporting our products from the country in which they are manufactured to our customers. Further changes to or our failure to comply with these and similar regulations could (1) restrict our ability to expand, build or acquire new facilities, (2) require us to acquire costly control equipment, (3) cause us to incur expenses associated with remediation of contamination, (4) cause us to modify our manufacturing or shipping processes or (5) otherwise increase our cost of doing business, which may have a negative impact on our financial condition, results of operations and cash flows. In addition, the potential adoption of new laws, rules or regulations related to climate change and the use or sale of PFAS-containing products poses risks, including subjecting us to future costs and liabilities, that could harm our results of operations or affect the way we conduct our businesses. For example, new or modified regulations could require us to make substantial expenditures to enhance our environmental compliance efforts.

We are exposed to various risks from our regulatory environment.

We are subject to risks related to new, different, inconsistent, or even conflicting laws, rules, and regulations that may be enacted by legislative or executive bodies and/or regulatory agencies in the countries where we operate; disagreements or disputes related to international trade; and the interpretation and application of laws, rules, and regulations. As a public company with global operations, we are subject to the laws of multiple jurisdictions and the rules and regulations of various governing bodies, including those related to health and safety, export controls, financial and other disclosures, corporate governance, privacy, anti-corruption, such as the Foreign Corrupt Practices Act and other local laws prohibiting corrupt payments to governmental officials or customers, conflict minerals or other social responsibility legislation, employment practices, immigration or travel regulations and antitrust regulations, among others. Each of these laws, rules and regulations imposes costs on our business, including financial costs and potential diversion of our management's attention, and may present risks to our business, including potential fines, restrictions on our actions and reputational damage if we do not fully comply. The volume of changes to such laws, rules and regulations may increase in the countries where we operate. To maintain high standards of corporate governance and public disclosure, we intend to invest in appropriate resources to comply with evolving standards. Changes in or ambiguous interpretations of laws, regulations and standards may create uncertainty regarding compliance matters. Efforts to comply with new and changing regulations have resulted in, and are likely to continue to result in, increased administrative expenses and diversion of management's time and attention from revenue-generating activities to compliance activities. If we are found by a court or regulatory agency not to be in compliance with laws and regulations, our reputation, business, financial condition and/or results of operations could be adversely affected, we may be disqualified or barred from participating in certain activities and we may be forced to modify our operations to achieve full compliance.

Changes in taxation or adverse tax rulings could adversely affect our results of operations.

We operate in many foreign countries and are subject to taxation at various rates and audit by multiple taxing authorities. Our results of operations could be affected by tax audits, changes in tax rates, changes in laws and regulations governing the calculation, location and taxation of earned profit, changes in laws and regulations affecting our ability to realize deferred tax assets on our balance sheet and changes in laws and regulations relating to the repatriation of cash into the United States. Each quarter, we forecast our tax liability based on our forecast of our performance for the year in each tax jurisdiction. If our performance forecast changes, our forecasted tax liability would also likely change, perhaps materially. We have undertaken and expect to continue to undertake a number of complex internal reorganizations of our foreign subsidiaries in order to rationalize and streamline our foreign operations, focus our management efforts on certain local opportunities and take advantage of favorable business conditions in certain localities. These or any future reorganizations could result in adverse tax consequences in one or more jurisdictions, which could adversely impact our profitability from foreign operations and result in a material reduction in our results of operations. Various other jurisdictions, including members of the Organization for Economic Cooperation and Development, are considering changes to their tax laws, including provisions intended to address base erosion and profit shifting by taxpayers. Any tax reform adopted in these or other countries may exacerbate the risks described above.

Risks Related to Our Indebtedness

We have a substantial amount of indebtedness and may in the future incur substantially more debt, each of which could adversely affect our ability to obtain financing in the future and react to changes in our business.

As of January 5, 2024, we had an aggregate principal amount of $6.11 billion of indebtedness outstanding, including the $1.82 billion from our senior secured term loan facility due 2029 (the “Term Loan Facility”), $2.08 billion aggregate principal amount of the 4.78% senior secured notes due April 20, 2029, $2.21 billion aggregate principal amount of the 5.98% senior unsecured notes due June 20, 2030, our 4.375% senior unsecured notes due April 20, 2028, and our 3.625% senior unsecured notes due May 6, 2029 (collectively, the “Notes”). In addition, we have approximately $747.5 million of unutilized capacity under our senior secured revolving credit facility due 2027 (the “Revolving Facility”). We refer to the Term Loan Facility and the Revolving Facility as the “Credit Facilities”. The credit agreements that govern the Credit Facilities are referred to collectively as the “Amended Credit Agreement”. Further, we may incur significant additional secured and unsecured indebtedness in the future. Although the indentures governing the Notes (the “Indentures”), and the Amended Credit Agreement restrict our ability to incur additional indebtedness, the restrictions have a number of significant qualifications and exceptions. For example, the Amended Credit Agreement provides that we can request additional loans and commitments up to the greater of $1.43 billion or 100% of our EBITDA, as well as additional amounts if our secured net leverage ratio is less than a specified ratio. Further, these restrictions do not prevent us from incurring monetary obligations that do not constitute indebtedness. If we add new indebtedness and other monetary obligations to our current debt levels, the related risks that we now face would intensify. Our debt could have important consequences, including:

• limiting

our ability to obtain additional financing to fund future working capital, capital expenditures, acquisitions or other general corporate purposes;

• requiring a substantial portion of our cash flow to be dedicated to debt service payments instead of other purposes;

• increasing our vulnerability to adverse changes in general economic, industry and competitive conditions;

• exposing

us to increased interest expense for borrowings with variable interest rates, including borrowings under the Credit Facilities; and

• placing

us at a disadvantage compared to other, less leveraged competitors or competitors with comparable debt having more favorable terms.

We may be unable to generate sufficient cash to service our indebtedness and may be forced to take other actions, which may not be successful, to satisfy our obligations under our indebtedness.

We may be unable to maintain sufficient cash flow from operating activities to permit us to pay the principal of, premium, if any, and interest on our indebtedness. Our ability to make scheduled payments on or to refinance our debt obligations depends on our financial condition and operating performance and the condition of the capital markets, which are subject to prevailing economic, industry and competitive conditions, as well as many financial, business, legislative, political, regulatory and other factors beyond our control. If our cash flow and capital resources are insufficient to fund our debt service obligations, we could face substantial liquidity problems, be forced to reduce or delay investments and capital expenditures, dispose of material assets or operations, seek additional debt or equity capital or restructure or refinance our indebtedness, any of which could have a material adverse effect on our business, financial position and results of operations. In addition, the level and quality of our earnings, operations, business and management, among other things, will impact the determination of our credit ratings. A decrease in the ratings assigned to us by the ratings agencies may negatively impact our access to the debt capital markets and increase our cost of borrowing. In addition, we may be unable to maintain the current credit worthiness or prospective credit rating of the Company. Any actual or anticipated changes or downgrades in such credit rating may have a negative impact on our liquidity, capital position or access to capital markets and affect our ability to obtain any future required financing on acceptable terms or at all. Any refinancing of our debt could be at higher interest rates and may require us to comply with more onerous covenants, which could further restrict our business operations. We may not be able to implement any refinancing on commercially reasonable terms or at all and, even if successful, a refinancing may not allow us to meet our scheduled debt service obligations. The agreements governing our indebtedness restrict our ability to dispose of assets and use the proceeds of such dispositions, and we may be unable to consummate any dispositions or generate proceeds sufficient to meet our debt service obligations. If we cannot make scheduled payments on our debt, holders of the Notes and lenders under the Credit Facilities could declare all outstanding principal and interest to be due and payable, the lenders under the Revolving Facility could terminate their commitments to advance further loans, our secured lenders could foreclose against the assets securing their borrowings and we could be forced into bankruptcy or liquidation.

The terms of the Amended Credit Agreement and the Indentures may restrict our operations, particularly our ability to respond to changes or raise additional funds.

The Amended Credit Agreement contains restrictive covenants that impose significant operating and financial restrictions that may limit our and our restricted subsidiaries' ability to take actions that may be in our long-term best interest, including restrictions on our and our restricted subsidiaries' ability to:

- incur additional indebtedness and guarantee indebtedness;
- pay dividends or make other distributions in respect of, or repurchase or redeem, capital stock;
- prepay, redeem or repurchase certain debt;
- make investments, loans, advances and acquisitions;
- engage in sale-leaseback or hedging transactions;
- create liens on, sell or otherwise dispose of assets, including capital stock of our subsidiaries;
- enter into transactions with affiliates;
- enter into agreements that restrict the ability to create liens, pay dividends or make loan repayments;
- alter the businesses we conduct; and
- merge or sell all or substantially all of our assets or incur a change of control in our capital stock ownership. Also,

the Indentures contain limited covenants, such as a covenant restricting our ability and certain of our subsidiaries' ability to incur certain debt secured by liens, engage in sale-leaseback and incur additional indebtedness by any restricted subsidiary. In addition, the restrictive covenants under the credit agreement governing the Revolving Facility may, depending on the amount of revolving borrowings, unreimbursed letter of credit drawings and undrawn letters of credit, require us to maintain a secured net leverage ratio, which we may be unable to meet. Our failure to comply with these covenants could result in the acceleration of some or all of our indebtedness, which could lead to bankruptcy, reorganization or insolvency.

Risks Related to Owning our Common Stock

The price of our common stock has been and may remain volatile.

The price of our common stock has been volatile. In 2023, the closing price of our stock on The Secondary Stock Exchange Global Select Market ("Secondary Stock Exchange") ranged from a low of $83.37 to a high of $158.08, and, as in past years, the price of our common stock may show even greater volatility in the future. The trading price of our common stock is subject to significant volatility in response to numerous factors, many of which are beyond our control or may be unrelated to our operating results, including the following:

- changes to our financial guidance, as well as potential decreased confidence in any guidance we do provide;

- changes

in global economic and geopolitical conditions, including those resulting from trade tensions, rising inflation, and fluctuations in foreign currency exchange and interest rates;

- failure to meet the expectations of securities analysts, which may vary significantly from our actual results;
- changes in financial estimates by securities analysts;
- press releases or announcements by, or changes in market values of, comparable companies;
- high volatility in price and volume in the markets for high-technology stocks;
- public perception of equity values of publicly traded companies;
- fluctuations in our results of operations; and
- other risks and uncertainties described in this Annual Report on Annual Statutory Report and in our other filings with the Financial Oversight Commission. Fluctuations

in our results of operations could cause our stock price to decline significantly. We believe that period-to-period comparisons of our results of operations may not be meaningful, and you should not rely upon them as indicators of our future performance. Future decreases in our stock price may adversely impact our ability to raise sufficient additional capital in the future, if needed.

We may decrease or discontinue cash dividends and may never adopt a new program to repurchase our shares of common stock.

Future payments of quarterly dividends and any future repurchases of shares of our common stock are subject to capital availability and periodic determinations by our Board of Directors that they are in the best interest of our stockholders and comply with all laws and applicable agreements. Future dividends and any future share repurchases may be affected by, among other factors, potential capital requirements for acquisitions and the funding of our research and development activities; legal risks; changes in federal and state income tax laws or corporate laws; contractual restrictions, such as financial or operating covenants in our debt arrangements; availability of domestic cash flow; and changes to our business model. The amounts of our dividend payments may change from time to time, and we may decide at any time to reduce, suspend or discontinue the payment of dividends or the repurchase of shares. A reduction, suspension or discontinuation of our dividend payments or the cessation of our share repurchase program could have a negative effect on the price of our common stock and may harm our reputation.

Provisions in our charter documents and Delaware law may delay or prevent an acquisition of us, which could decrease the value of our shares.

Our certificate of incorporation, our by-laws and Delaware law contain provisions that could make it harder for a third party to acquire us without the consent of our board of directors. These provisions include limitations on actions by written consent of our stockholders. Our certificate of incorporation makes us subject to the anti-takeover provisions of Section 203 of the Delaware General Corporation Law. In general, Section 203 prohibits publicly held Delaware corporations from engaging in a “business combination” with an “interested stockholder” for a period of three years after the date of the transaction in which the person became an interested stockholder, unless the business combination is approved in a prescribed manner. This provision could discourage parties from bidding for our shares of common stock and could, as a result, reduce the likelihood of an increase in the price of our common stock that would otherwise occur if a bidder sought to buy our common stock. Our certificate of incorporation authorizes our Board of Directors to issue, without further stockholder approval, up to 5,000,000 shares of preferred stock in one or more series and to fix and designate the rights, preferences, privileges and restrictions of the preferred stock, including dividend rights, conversion rights, voting rights, redemption rights and liquidation preferences. The holders of any shares of preferred stock could have preferences over the holders of our common stock with respect to dividends and liquidation rights. Any issuance of preferred stock may have the effect of delaying, deterring or preventing a change in control. Any issuance of preferred stock could decrease the amount of earnings and assets available for distribution to the holders of common stock and could adversely affect the rights and powers, including voting rights, of the holders of common stock. The issuance of preferred stock could have the effect of decreasing the market price of our common stock.

General Risks

Competition from new or existing companies could harm our financial condition, results of operations and cash flow.

We operate in a highly competitive, global industry. We face many domestic and international competitors, some of which have substantially greater manufacturing, financial, research and development and marketing resources than we do. In addition, some of our competitors may have better-established customer relationships than we do, which may enable them to have their products specified for use more frequently and more quickly by these customers. We also face competition from smaller, regional companies that focus on serving customers in their regions. Further, customers continually evaluate the benefits of internal manufacturing versus outsourcing, and a customer’s decision to internally manufacture products that we provide may negatively impact us. If we are unable to maintain our competitive position, we could experience downward pressure on prices, fewer customer orders, reduced margins, the inability to take advantage of new business opportunities and a loss of market share, any of which could have a material adverse effect on our results of operations. Further, we expect that existing and new competitors will improve their products and introduce new products with enhanced performance characteristics. The introduction of new products or more efficient production of existing products by competitors could diminish our market share and increase pricing pressure on our products. Our competitors include companies outside of the U.S., including companies in countries where foreign governments seek to build a domestic-centric semiconductor ecosystem. From time to time, governments around the world may provide incentives or make other investments that could benefit and give competitive advantages to our competitors. For example, in August 2022, the U.S. government enacted the CHIPS and Science Act of 2022 (the “CHIPS Act”) to provide financial incentives to the U.S. semiconductor industry. Government incentives, including any that may be offered in connection with the CHIPS Act, may not be available to us on acceptable terms or at all. If our competitors can benefit from such government incentives and we cannot, it could strengthen our competitors' relative position and have a material adverse effect on our business.

We may acquire other businesses, form joint ventures or divest businesses, any of which could negatively affect our financial performance.

We intend to continue to engage in business combinations, acquisitions, joint ventures, investments, divestitures or other types of collaborations to (1) address gaps in our product offerings, (2) adjust our business and product portfolio to meet our ongoing strategic objectives, (3) diversify into new and complementary markets, (4) increase our scale or (5) accomplish other strategic objectives. These transactions involve numerous risks to our business, financial condition and operating results, including but not limited to:

• difficulty

in identifying suitable acquisition candidates and completing transactions at appropriate valuations, in a timely manner, on a cost-effective basis or at all, due to substantial competition for acquisition targets;

• inability to successfully integrate any acquired businesses into our business operations;
• failure to realize the anticipated synergies or other benefits of any such transaction;
• entry into markets in which we have limited or no prior experience;
• finding acquirors and obtaining adequate value for businesses that no longer meet our strategic objectives;
• difficulties

surrounding the disentanglement of a divested business, including the diversion of resources away from our business operations to address such matters;

• inability

to complete proposed or pending transactions due to factors such as the failure or inability to obtain regulatory or other approvals, which may be exacerbated by the recent, more aggressive regulatory approaches to merger control globally, such as the July 24, 2023 joint statement of antitrust policy by the Department of Justice and Federal Trade Commission and the April 20, 2023 Provisions on the Review of Concentrations of Undertakings issued by China's State Administration for Market Regulation, among others;

• requirements

imposed by government regulators in connection with their review of a transaction, which may include, among other things, divestitures and restrictions on the conduct of our existing business or the acquired business;

• undertaking

multiple transactions at the same time in order to take advantage of acquisition or divestiture opportunities that do arise, which could strain our ability to effectively execute and integrate such transactions;

• diversion

of management's attention from our day-to-day business due to dedication of significant management resources to such transactions;

• employee uncertainty and lack of focus during the integration process that may also disrupt our business;
• risk of litigation or claims associated with a proposed or completed transaction;
• challenges

associated with managing new, more diverse and more widespread operations, projects and people, potentially located in regions where we have not historically conducted or operated our business;

• dependence on unfamiliar or less secure supply chains and inefficient scale of the acquired entity;
• increasing costs of performing due diligence to meet the expectations of investors and government regulators;
• despite

our due diligence, we could assume unknown, underestimated or contingent liabilities, such as potential environmental, health and safety liabilities, any of which could lead to costly litigation or mitigation actions;

• an

acquired technology or product may have inadequate or invalid intellectual property protection or may be subject to claims of infringement by a third party, which may result in claims for damages and lower than anticipated revenue;

• negative

effects on our reported results of operations from dilutive results from operations and/or from future potential impairment of acquired assets, including goodwill, related to acquisitions;

• an

acquired company may have inadequate or ineffective internal controls over financial reporting, disclosure controls and procedures, cybersecurity, privacy, environmental, health and safety, anti-bribery, anti-corruption, human resource or other policies or practices, which may require unexpected or additional integration, mitigation and remediation costs;

- reductions

in cash or increases in debt to finance transactions, which reduce the cash flow available for general corporate or other purposes, including share repurchases and dividends; and

- difficulties in retaining key employees or customers of an acquired business.

Climate change may have a long-term impact on our business.

There are inherent climate-related risks wherever our business is conducted. Changes in market dynamics, stakeholder expectations, local, national and international climate change policies, and the frequency and intensity of extreme weather events on critical infrastructure in the United States and abroad, all have the potential to disrupt our business and operations. Such events could result in a significant increase in our costs and expenses and harm our future revenue, cash flows and financial performance. Global climate change is resulting in, and may continue to result, in certain natural disasters and adverse weather events, such as drought, wildfires, storms, sea-level rise and flooding, occurring more frequently or with greater intensity, which could cause business disruptions and impact employees' abilities to commute or to work from home effectively.

Item 7: Management's Discussion and Analysis of Financial Condition and Results of Operations.

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations.

The following discussion and analysis of the Company's consolidated financial condition and results of operations should be read along with the consolidated financial statements and the accompanying notes thereto included elsewhere in this Annual Report on Annual Statutory Report. This discussion contains forward-looking statements that involve numerous risks and uncertainties, including, but not limited to, those described in Item 1A, "Risk Factors" and the "Cautionary Statements" section of this Item 7 below. You should review Item 1A "Risk Factors" of this Annual Report on Annual Statutory Report for a discussion of important factors that could cause actual results to differ materially from the results described in or implied by the forward-looking statements contained in the following discussion and analysis. The Company has elected to omit discussion of the earliest of the three years covered by the consolidated financial statements presented except for the segment analysis. Information pertaining to fiscal year 2021 results of operations and the year-over-year comparison of changes in our Financial Condition and Results of Operations as of and for the year ended January 5, 2023 and 2021 can be found in Part II, Item 7, "Management's Discussion and Analysis of Financial Condition and Results of Operations" of our Annual Report on Annual Statutory Report for the year ended January 5, 2023, filed on February 28, 2023.

Cautionary Statements

This Annual Report on Annual Statutory Report and the portions of the Company's Definitive Governance Disclosure Document incorporated by reference in this Annual Report on Form 10-K contain "forward-looking statements." The words "believe," "expect," "anticipate," "intend," "estimate," "forecast," "project," "should," "may," "will," "would" or the negative thereof and similar expressions are intended to identify such forward-looking statements. These forward-looking statements may include statements about supply chain matters; inflationary pressures; future period guidance or projections; the Company's performance relative to its markets, including the drivers of such performance; market and technology trends, including the duration and drivers of any growth trends; the development of new products and the success of their introductions; the focus of the Company's ER&D projects; the Company's ability to execute on our business strategies, including with respect to the Company's expansion of its manufacturing presence in Taiwan and in Henderson; the Company's capital allocation strategy, which may be modified at any time for any reason, including share repurchases, dividends, debt repayments and potential acquisitions; the impact of the acquisitions and divestitures the Company has made and commercial partnerships the Company has established, including the acquisition of Quantum Materials Division (now known as Quantum Materials Division LLC) ("Quantum Materials Division"); trends relating to the fluctuation of currency exchange rates; future capital and other expenditures, including estimates thereof; the Company's expected tax rate; the impact, financial or otherwise, of any organizational changes; the impact of accounting pronouncements; quantitative and qualitative disclosures about market risk; and other matters. These forward-looking statements are based on current management expectations and assumptions only as of the date of this Annual Report, are not guarantees of future performance and involve substantial risks and uncertainties that are difficult to predict and that could cause actual results to differ materially from the results expressed in, or implied by, these forward-looking statements. These risks and uncertainties include, but are not limited to, weakening of global and/or regional economic conditions, generally or specifically in the semiconductor industry, which could decrease the demand for the Company's products and solutions; the level of, and obligations associated with, the Company's indebtedness, including the debts incurred in connection with the acquisition of Quantum Materials Division; risks related to the acquisition and integration of Quantum Materials Division, including unanticipated difficulties or expenditures relating thereto, the ability to achieve the anticipated synergies and value-creation contemplated by the acquisition of Quantum Materials Division and the diversion of management time on transaction-related matters; raw material shortages, supply and labor constraints, price increases, inflationary pressures and rising interest rates; operational, political and legal risks of the Company's international operations; the Company's dependence on sole source and limited source suppliers; the Company's ability to meet rapid demand shifts; the Company's ability to continue technological innovation and introduce new products to meet customers' rapidly changing requirements; substantial competition; the Company's concentrated customer base; the Company's ability to identify, complete and integrate acquisitions, joint ventures, divestitures or other similar transactions; the Company's ability to effectively implement any organizational changes; the Company's ability to protect and enforce intellectual property rights; the impact of regional and global instabilities, hostilities and geopolitical uncertainty, including, but not limited to, the ongoing conflicts between Ukraine and Russia, between Israel and Hamas and the current conflict in the Red Sea, as well as the global responses thereto; the increasing complexity of certain manufacturing processes; changes in government regulations of the countries in which the Company operates, including the imposition of tariffs, export controls and other trade laws and restrictions and changes to national security and international trade policy, especially as they relate to China; fluctuation of currency exchange rates; fluctuations in the market price of the Company's stock; and other matters. These forward-looking statements are based on current management expectations and assumptions only as of the date of this Annual Report on Annual Statutory Report. They are not guarantees of future performance and involve substantial risks and uncertainties that are difficult to predict and that could cause actual results to differ materially from the results expressed in, or implied by, these forward-looking statements. These risks and uncertainties include, but are not limited to, the risk factors and additional information described in this Annual Report on Annual Statutory Report under the caption "Risk Factors," elsewhere in this Annual Report on Annual Statutory Report and in the Company's other periodic filings. Except as required under the federal securities laws and the rules and regulations of the Financial Oversight Commission, the Company undertakes no obligation to update publicly any forward-looking statements or information contained herein, which speak as of their respective dates.

Overview

This overview is not a complete discussion of the Company's financial condition, changes in financial condition and results of operations; it is intended merely to facilitate an understanding of the most salient aspects of its financial condition and operating performance and to provide a context for the detailed discussion and analysis that follows, and must be read in its entirety in order to fully understand the Company's financial condition and results of operations.

The Company is a leading supplier of mission-critical advanced materials and process solutions for the semiconductor and other high-technology industries. We leverage our unique breadth of capabilities to help our customers improve their productivity, performance and technology in the most advanced manufacturing environments. In the third quarter of 2023, the Company realigned its segments into three reportable segments discussed below, which align with the key elements of the advanced semiconductor manufacturing ecosystem. The current annual and succeeding annual periods will disclose the reportable segments with prior periods recast to reflect the change.

• The

Materials Solutions segment, or MS, provides materials-based solutions, such as chemical mechanical planarization ("CMP") slurries and pads, deposition materials, process chemistries and gases, formulated cleans, etchants and other specialty materials that enable our customers to achieve better device performance and faster time to yield, while providing for lower total cost of ownership. ■ The Microcontamination Control segment, or MC, offers advanced solutions that improve customers' yield, device reliability and cost by filtering and purifying critical liquid chemistries and gases used in semiconductor manufacturing processes and other high-technology industries. ■ The Advanced Materials Handling segment, or AMH, develops solutions that improve customers' yields by protecting critical materials during manufacturing, transportation, and storage, including products that monitor, protect, transport and deliver critical liquid chemistries, wafers, and other substrates for a broad set of applications in the semiconductor, life sciences and other high-technology industries. These segments share common business systems and processes, technology centers and technology roadmaps. With the complementary capabilities across these segments, we believe we are uniquely positioned to create new, co-optimized and increasingly integrated solutions for our customers, which should translate into improved device performance, lower cost of ownership and faster time to market. This capability has been further enhanced with the acquisition of Quantum Materials Division. For example, we can now develop and provide complementary offerings solving customers' complex manufacturing challenges across the deposition, CMP process and post-CMP modules with co-optimized products from each of our divisions, such as advanced deposition materials, CMP slurries, pads and post-CMP cleaning chemistries from our MS segment, CMP slurry filters from our MC segment, and CMP slurry high-purity packaging and fluid monitoring systems from our AMH segment.

Impact of New Export Control Regulations

On October 22, 2023, the U.S Department of Commerce, Bureau of Industry and Security ("BIS") announced updates to export control regulations, originally issued on October 12, 2022, regarding the sale of certain products and services related to advanced computing items, semiconductor manufacturing equipment, and items that can support end uses related to the development and production of advanced-node integrated circuits and semiconductor manufacturing equipment, among others. The updated rules modify and expand restrictions on the sale of products and the provision of certain services by U.S. persons to some companies and domestic fabs located in certain countries, including China, without prior U.S. governmental authorization. See Item 1A, "Risk Factors," in this Annual Report on Annual Statutory Report for additional information regarding risk associated with the impact of new export control regulations, including under the caption "Tariffs, export controls and other trade laws and restrictions resulting from international trade disputes, strained international relations and changes to foreign and national security policy, especially as they relate to China, could have an adverse impact on our operations and reduce the competitiveness or availability of our products relative to local and global competitors."

Impact of Conflicts in the Middle East

The military conflict between Israel and militant groups led by Hamas and the current conflict in the Red Sea have caused uncertainty in the global markets, including, but not limited to, disruptions to shipping routes. Revenue relating to products manufactured from raw materials or components sourced from or through this region does not constitute a material portion of our business and historically we have not had significant revenue in this region. There continues to be uncertainty regarding the ultimate impact these conflicts will have on the global economy, supply chains, logistics, fuel prices, raw material pricing and our business.

Recent Events

On February 15, 2023, the Company terminated the definitive agreement with Lubrex Chemicals to sell its PIM business. At the time of the termination, the transaction had not received clearance under the HSR Act. In accordance with the terms of the definitive agreement, the Company received a $15.6 million termination fee from Lubrex Chemicals in the first quarter of 2023. Also in the first quarter of 2023, the Company incurred a fee of $1.43 million to the third-party financial adviser it had engaged to assist with the transaction. The assets and liabilities of the PIM business are classified as held for sale as of January 5, 2024. See Note 5 to our consolidated financial statements for further discussion. On March 6, 2023, the Company completed its divestiture of the PPI business. The Company received proceeds of $174.59 million. See Note 5 to our consolidated financial statements for further discussion. On March 15, 2023 and September 16, 2023, the Company amended its Existing Credit Agreement. The First Amendment, dated March 15, 2023, provided for, among other things, the refinancing of the Company's then-outstanding term loans B under the senior secured term loan facility due 2029 in an aggregate principal amount of $3.24 billion with a new tranche of term loans B in an aggregate principal amount of $3.24 billion. The amended loans under the First Amendment bore interest at a rate per annum equal to the Secured Overnight Financing Rate ("SOFR") plus an applicable margin of 2.75% which was a reduction from the applicable margin of 3.00% prior to the First Amendment. The Second Amendment, dated September 16, 2023, provides for, among other things, the refinancing of the Company's outstanding term loans B under the senior secured term loan facility due 2029 in an aggregate principal amount of $3.01 billion with a new tranche of term loans B in an aggregate principal amount of $3.01 billion. The outstanding term loans B under the amended senior secured term loan facility due 2029 bear interest, at a rate per annum equal to the SOFR plus an applicable margin of 2.50% which is a reduction from the applicable margin of 2.75% provided for in the First Amendment. See Note 11 to our consolidated financial statements for further discussion. On June 10, 2023, the Company announced the termination of an Alliance Agreement (the "Alliance Agreement") between the Company and Veralux Industrial Chemistry Inc., a global business unit of Veralux Materials Group. ("Veralux Industrial Chemistry"). In connection with the termination of the Alliance Agreement, Zava Technologies received net proceeds of $248.56 million in 2023. See Note 5 to our consolidated financial statements for further discussion. On October 7, 2023, the Company completed its divestiture of the Electronic Chemicals ("EC") business. The Company received cash proceeds of $958.23 million, or net proceeds of $877.76 million. See Note 5 to our consolidated financial statements for further discussion.

Critical Accounting Policies and Estimates

Management's discussion and analysis of financial condition and results of operations are based upon the Company's consolidated financial statements, which have been prepared in accordance with accounting principles generally accepted in the United States ("generally accepted accounting principles"). The preparation of these consolidated financial statements requires the Company to make estimates, assumptions and judgments that affect the reported amounts of assets, liabilities, revenues and expenses and related disclosure of contingent assets and liabilities. At each balance sheet date, management evaluates its estimates, including, but not limited to, those related to long-lived assets (property, plant and equipment, and identified intangible assets), goodwill and income taxes. The Company bases its estimates on historical experience and various other assumptions that management believes to be reasonable under the circumstances. Management's utilization of different judgments or estimates could result in material differences in the amount and timing of the Company's results of operations for any period. In addition, actual results could be different from the Company's current estimates, possibly resulting in increased future charges to earnings. Our critical accounting policies that are most significantly affected by estimates, assumptions and judgments used in the preparation of the Company's consolidated financial statements relate to business acquisitions and are discussed below. See Note 1 to the Company’s consolidated financial statements for additional information about the Company’s other significant accounting policies.

Impairment of Goodwill

Goodwill is tested for impairment annually as of August 31. If circumstances change during interim periods between annual tests that would more likely than not reduce the fair value of a reporting unit below its carrying value, the Company will test goodwill for impairment. Factors that would necessitate an interim goodwill impairment assessment include a sustained decline in the Company's stock price, effects on a reporting unit such as a change in the composition or carrying amounts of its net assets, prolonged negative industry or economic trends, or significant under-performance relative to expected, historical or projected future operating results. Management uses judgment to determine whether to use a qualitative analysis or a quantitative fair value measurement for its goodwill impairment testing. The Company's fair value measurement approach combines the income and market valuation techniques for each of the Company’s reporting units that carry goodwill. These valuation techniques use estimates and assumptions including, but not limited to, the determination of appropriate market comparable, projected future cash flows (including timing and profitability), the discount rate reflecting the risk inherent in future cash flows, the perpetual growth rate, and projected future economic and market conditions. If a reporting unit fails the quantitative impairment test, impairment expense is immediately recorded as the difference between the reporting unit’s fair value and carrying value not to exceed the amount of goodwill recorded.

Business Acquisitions

The Company accounts for acquired businesses using the acquisition method of accounting, which requires that the assets acquired and liabilities assumed be recorded at the date of acquisition at their respective fair values. The judgments made in determining the estimated fair value assigned to each class of assets acquired and liabilities assumed, as well as asset lives, can materially impact net income. Accordingly, for significant items, the Company typically obtains assistance from a third-party valuation firm. There are several methods that can be used to determine the fair value of assets acquired and liabilities assumed in a business combination. For intangible assets, the Company normally utilizes the “income method,” which starts with a forecast of all of the expected future net cash flows attributable to the subject intangible asset. These cash flows are then adjusted to present value by applying an appropriate discount rate that reflects the risk factors associated with the cash flow streams. Depending on the asset valued, the key assumptions included one or more of the following: (1) future revenue growth rates, (2) future gross margin, (3) future selling, general and administrative expenses, (4) royalty rates, and (5) discount rates. Estimating the useful life of an intangible asset also requires judgment. For example, different types of intangible assets will have different useful lives, influenced by the nature of the asset, competitive environment, and rate of change in the industry. Certain assets may even be considered to have indefinite useful lives. All of these judgments and estimates can significantly impact the determination of the amortization period of the intangible asset, and thus net income.

Results of Operations

Year ended January 5, 2024 compared to year ended January 5, 2023

The following table sets forth the results of operations and the relationship between various components of operations, stated as a percent of net sales, for 2023 and 2022.

Net sales

For 2023, net sales were $4581.07 million, increased by $314.47 million, or 7%, from 2022. An analysis of the factors underlying the increase in net sales is presented in the following table:

The sales increase was attributable, in part, to a total of $699.14 million of sales of new products resulting from the acquisition of QUANTUM Materials. This increase was partially offset by an absence of sales totaling $151.06 million resulting from the divestitures of PPI and the Electronic Chemicals business as well as the termination of the Alliance Agreement. Total net sales also reflected unfavorable foreign currency translation effects of $43.16 million, mainly due to the significant weakening of the Japanese yen relative to the U.S. dollar, and decreased demand from customers in the semiconductor market, resulting in a decrease of $190.45 million compared to the year ago period ended January 5, 2023.

Sales percentage on a geographic basis for 2023 and 2022 and the percentage increase (decrease) in sales for 2023 compared to sales for 2022 were as follows:

The increase in sales to customers for all countries and regions, except Taiwan, in the table above was principally driven by the inclusion of sales from the acquisition of Quantum Materials Division. The decrease in sales in Taiwan primarily relates to lower sales demand of AMH and MC products partially offset by an increase in sales resulting from the inclusion of sales from the Quantum Materials Division acquisition.

Gross margin

The following table sets forth gross margin as a percentage of net sales:20232022Percentage point changeGross margin as a percentage of net sales:42.5 %42.5 %— Gross margin was unchanged for 2023 compared to 2022, which was primarily due to the absence of a $80.47 million charge for fair value write-up of acquired Quantum Materials Division inventory sold during the year ended January 5, 2023, partially offsetting the effect of lower plant utilization in 2023.

Selling, general and administrative expenses

Selling, general and administrative (“SG&A”) expenses consist primarily of payroll and related expenses for the sales and administrative staff, professional fees (including accounting, legal and technology costs and expenses), and sales and marketing costs. SG&A expenses for 2023 increased $42.51 million, or 6%, to $749.06 million from $706.55 million in 2022.

An analysis of the factors underlying the increase in SG&A expenses is presented in the following table:

Engineering, research and development expenses

ER&D expenses consist of expenses for the support of current product lines and the development of new products and manufacturing technologies. These expenses were $360.49 million in 2023 and $297.7 million in 2022.

An analysis of the factors underlying the increase in ER&D expenses is presented in the following table:

The Company’s overall ER&D efforts will continue to focus on developing and improving its technology platforms for semiconductor and advanced processing applications and identifying and developing products for new applications. The Company often works directly with its customers to address their particular needs. Amortization of intangible assets Amortization of intangible assets was $278.85 million in 2023 compared to $187.2 million for 2022. The increase primarily reflects the full-year of amortization expense associated with the intangible assets acquired in the Quantum Materials Division acquisition. Goodwill impairment The Company recorded a goodwill impairment charges of $149.76 million in 2023. See Note 3 to the Company’s consolidated financial statements for further discussion. Gain on termination of Alliance Agreement In connection with the termination of the Alliance Agreement, the Company recognized a pre-tax gain, net of $240.24 million in 2023. See Note 5 to the Company’s consolidated financial statements for further discussion. Interest expense Interest expense was $406.12 million in 2023 and $276.51 million in 2022. Interest expense includes interest associated with debt outstanding and the amortization of debt issuance costs associated with such borrowings. The increase was primarily driven by a full year of interest expense based on timing on Quantum Materials Division close in 2022. Interest income Interest income was $14.69 million in 2023 and $4.81 million in 2022. The increase reflects rising average interest rates and cash balances. Other expense, net Other expense, net, was $33.02 million in 2023 compared to $31.07 million in 2022. In 2023, other expense, net consisted mainly of loss of extinguishment and modification of debt of $38.87 million associated with the repayments on the Company's bridge credit facility and senior secured term loan facility and the amendments of the Company's Existing Credit Agreement (see Note 11 to the Company's consolidated financial statements) and foreign currency transaction losses of $7.41 million, partially offset by net proceeds received of $14.17 million resulting from the termination of the definitive agreement with Lubrex Chemicals related to the PIM business. In 2022, other expense, net consisted mainly of consisted mainly of foreign currency transaction losses of $29.9 million. Income tax expense The Company recorded income tax benefit of $10.92 million in 2023 compared to income tax expense of $49.66 million in 2022. The Company's effective tax rate was (4.9)% in 2023 compared to an effective tax rate of 15.4% in 2022. The decrease in the effective tax rate from 2022 to 2023 relates to a tax benefit of $22.62 million recorded in 2023 related to divestiture and impairment activity. Additionally, the tax rate was lower in 2023 due to changes to the jurisdictional income mix resulting from the acquisition of Quantum Materials Division and the temporary changes in the U.S. tax regulations pertaining to foreign tax credits. The tax rate in 2022 was higher due to non-deductible acquisition related costs and a decrease in discrete benefits related to share-based compensation. Net income Net income was $234.91 million, or $1.56 per diluted share, in 2023 compared to net income of $271.57 million, or $1.9 per diluted share, in 2022. The decrease reflects the Company's aforementioned operating results described in greater detail above. Non-generally accepted accounting principles Financial Measures Information The Company's consolidated financial statements are prepared in conformity with accounting principles generally accepted in the United States. The Company also utilizes certain non-generally accepted accounting principles financial measures as a complement to financial measures provided in accordance with generally accepted accounting principles in order to better assess and reflect trends affecting the Company's business and results of operations. See "Non-generally accepted accounting principles Information" included below in this section for additional detail, including the reconciliation of the Company's non-generally accepted accounting principles measures to the most directly comparable generally accepted accounting principles measures. The Company's non-generally accepted accounting principles financial measures include Adjusted EBITDA and Adjusted Operating Income, together with related percentage changes, and Non-generally accepted accounting principles Earnings Per Share, or EPS. Adjusted EBITDA decreased to $1225.12 million in 2023, compared to $1265.16 million in 2022. Adjusted EBITDA as a percent of net sales was 26.7% in 2023 compared to 29.7% in 2022. Adjusted Operating Income decreased by 8.1% to $1000.61 million in 2023, compared to $1089.27 million in 2022. Adjusted Operating Income as a percent of net sales was 21.8% in 2023 compared to 25.5% in 2022. Non-generally accepted accounting principles EPS decreased 29.2% to $3.43 in 2023, compared to $4.85 in 2022. The decreases in Adjusted EBITDA and Adjusted Operating Income are primarily due to lower plant utilization and higher operating expenses. The decrease in Non-generally accepted accounting principles EPS is primarily attributable to higher interest expense associated with debt financing in connection with the Quantum Materials Division acquisition.

Segment Analysis

In the third quarter of 2023, in order to align its segment financial reporting with a change in its business structure, the Company realigned its segments. Following the segment realignment, the Company's three reportable segments are as follows (1) Materials Solutions, (2) Microcontamination Control, and (3) Advanced Materials Handling. Accordingly, our segment information was restated retroactively in the third quarter of fiscal year 2023. The segment realignment had no impact on the Microcontamination Control or Advanced Materials Handling segment financial reporting. See Note 21 to the consolidated financial statements for additional information on the Company's three segments.

The following table and discussion reflects the results of operations of the Company's three reportable segments for the years ended January 5, 2024, 2022 and 2021.

Materials Solutions (MS)

For 2023, MS net sales increased to $2196.35 million, up 22% from $1794.26 million in 2022. The sales increase primarily reflects the inclusion of sales of product lines resulting from the acquisition of Quantum Materials Division totaling $699.14 million, partially offset by an absence of sales totaling $151.06 million resulting from the divestitures of PPI and the EC business, the termination of the Alliance Agreement and decline in demand across the semiconductor market.MS reported a segment profit of $385.32 million for 2023, up 35% compared to $284.96 million in 2022. The increase in MS's profit in 2023 was primarily associated with a gain of $240.24 million resulting from the termination of the Alliance Agreement (see Note 5 to our consolidated financial statements for further discussion), the absence of a $80.47 million charge for a fair value write-up resulting from the sale of acquired Quantum Materials Division inventory that was recorded in the year-ago period and segment profit attributed to the Quantum Materials Division acquisition, partially offset by a goodwill impairment charge of $136.24 million related to the EC reporting unit (see Note 3 to our consolidated financials statements for further discussion), a $11.57 million loss on asset held for sale related to the EC reporting unit, a goodwill impairment charge of $13.52 million, long-lived asset impairment charge of $39.65 million (see Note 3 to our consolidated financials statements for further discussion) and a $19.37 million loss on the sale of PPI. For 2022, MS net sales increased to $1794.26 million, up 94% from $924.69 million in 2021. The sales increase primarily reflects the inclusion of sales of $772.72 million attributed to acquisitions, primarily of QUANTUM Materials, and also reflects modestly improved sales of advanced deposition materials, formulated cleans, selective etch and specialty coating products. MS reported a segment profit of $284.96 million for 2022, up 31% compared to $218.14 million in 2021. The increase in MS's profit in 2022 was primarily due to the segment profit attributed to the Quantum Materials Division acquisition, partially offset by unfavorable product mix and a $80.47 million charge for a fair value write-up resulting from the sale of acquired Quantum Materials Division inventory.

Microcontamination Control (MC)

For 2023, MC net sales increased to $1465.88 million, up 2% from $1437.8 million in 2022. The sales increase was primarily due to improved sales from liquid filtration products.MC reported a segment profit of $513.89 million for 2023, down 4% compared to $534.95 million in 2022. The decrease in MC's profit in 2023 was primarily due to increased costs associated with the ramp up of our new manufacturing facility in Taiwan and increased investment in research and development. For 2022, MC net sales increased to $1437.8 million, up 20% from $1195.22 million in 2021. The sales increase was due to improved performance across substantially all platforms, with growth especially strong in liquid filtration and gas filtration products.MC reported a segment profit of $534.95 million for 2022, up 28% compared to $417.69 million in 2021. The increase in MC'sprofit in 2022 was primarily due to higher gross profit related to the increased sales volume, partially offset by higher operatingexpenses of 17%, primarily due to higher compensation costs.

Advanced Materials Handling (AMH)

For 2023, AMH net sales decreased 10% to $986.18 million from $1100.45 million in 2022. The sales decrease was mainly due to lower sales from our microenvironment solutions products.AMH reported a segment profit of $176.93 million for 2023, down 26% compared to $238.81 million in 2022. The decrease in AMH's profit in 2023 was primarily due to lower sales and lower factory utilization. For 2022, AMH net sales increased 20% to $1100.45 million from $916.37 million in 2021. The sales increase was mainly due to improved sales from wafer handling and fluid handling.AMH reported a segment profit of $238.81 million for 2022, up 15% compared to $208 million in 2021. The increase in AMH's profit in 2022 was primarily due to higher sales volume, partially offset by a 16% increase in operating expenses, primarily due to higher compensation costs.

Unallocated general and administrative expenses

Unallocated general and administrative expenses for 2023 totaled $148.46 million compared to $247.65 million for 2022. The $99.19 million decrease is primarily due to a $124.41 million decrease in deal, transaction and integration costs related to the acquisition of Quantum Materials Division, partially offset by an increase in employee costs of $18.33 million.

Liquidity and Capital Resources

We consider the following when assessing our liquidity and capital resources:

The Company has historically financed its operations and capital requirements through cash flow from its operating activities, long-term loans, lease financing and borrowings under domestic and international short-term lines of credit. Based on our analysis, we believe our existing balances of domestic cash and cash equivalents and our currently anticipated operating cash flows will be sufficient to meet our cash needs arising in the ordinary course of business for the next twelve months and for the longer term. We may seek to take advantage of opportunities to raise additional capital through additional debt financing or through public or private sales of securities. If in the future our available liquidity is not sufficient to meet the Company’s operating and debt service obligations as they come due, management would need to pursue alternative arrangements through additional equity or debt financing in order to meet the Company’s cash requirements. There can be no assurance that any such financing would be available on commercially acceptable terms, or at all. During 2023, we did not experience difficulty accessing capital and credit markets, but future volatility in the capital and credit markets may increase costs associated with issuing debt instruments or affect our ability to access those markets. In addition, it is possible that our ability to access the capital and credit markets could be limited at a time when we would like, or need, to do so, which could have an adverse impact on our ability to refinance maturing debt and/or react to changing economic and business conditions.

In summary, our cash flows for each period were as follows:

Operating activities

Cash provided by operating activities is net income adjusted for certain non-cash items and changes in assets and liabilities. Compared to 2022, the $360.49 million increase in cash provided by operating activities in 2023 was primarily driven by $375.57 million of changes in operating assets and liabilities, offset by a $15.21 million decrease of net income adjusted for non-cash reconciling items. Changes in operating assets and liabilities were driven by changes in trade accounts and notes receivable, inventories, accounts payable and accrued liabilities, income taxes payable and refundable income taxes. The change for trade receivables was mainly due to lower sales. The change for inventory was driven by the Company’s initiative to reduce inventory. The change for accounts payable and accrued liabilities was driven by lower vendor purchases and timing of payments. The change for income taxes payable and refundable income taxes was due to larger tax payments compared to the previous year.

Investing activities

Investing cash flows consist primarily of capital expenditures, cash used for acquisitions, proceeds from sales of businesses and proceeds from sales of property and equipment. In 2023, there was $719.03 million of cash provided by investing activities compared to $6429.41 million cash used in investing activities in 2022. The decrease resulted primarily from the absence of the prior-year acquisition of Quantum Materials Division, partially offset by proceeds from the sale of PPI and the EC businesses totaling $1059.5 million, net proceeds from the termination of the Alliance Agreement totaling $259.09 million in the current year, and a $12.22 million decrease in capital expenditures in the current year compared to the prior year. Acquisition of property and equipment totaled $593.84 million in 2023, which primarily reflected investments in facilities, equipment and tooling, compared to $606.06 million in 2022, which also primarily reflected investments in equipment and tooling. Capital expenditures in 2023 generally reflected more spending related to growth capacity, including our previously announced investment in our TPZ site and new facility in Henderson, Nevada. In 2022, the Company acquired Quantum Materials Division. The cash used to acquire these assets was $5817.37 million, net of cash acquired. The transaction is described in further detail in Note 4 to the Company’s consolidated financial statements.

Financing activities

Financing cash flows consist primarily of repurchases of common stock, payment of dividends to stockholders, issuance and repayment of short-term and long-term debt, and proceeds from the sale of shares of common stock through employee equity incentive plans. In 2023, there was $1667.38 million of cash used in financing activities compared to $6196.06 million cash provided by in financing activities in 2022. The change was primarily due to the net debt activity, which was a use of cash of $1637.61 million in 2023 compared to a source of cash of $6280.69 million. See Note 11 to the Company’s consolidated financial statements for further discussion of the debt financing that occurred during the year. The Company’s total dividend payments were $78.26 million in 2023 compared to $74.49 million in 2022. The Company has paid a cash dividend in each quarter since the fourth quarter of 2017. On January 22, 2024, the Company’s board of directors declared a quarterly cash dividend of $0.13 per share to be paid on February 26, 2024 to shareholders of record as of February 5, 2024.

Other Liquidity and Capital Resources Considerations

Debt at par value outstanding

On March 15, 2023, and September 16, 2023, the Company amended its Existing Credit Agreement. The First Amendment, dated March 15, 2023, provided for, among other things, the refinancing of the Company’s outstanding term loans B under the senior secured term loan facility due 2029 in an aggregate principal amount of $3.24 billion with a new tranche of term loans B in an aggregate principal amount of $3.24 billion. The amended loans under the First Amendment bore interest at a rate per annum equal to the Secured Overnight Financing Rate (“SOFR”) plus an applicable margin of 2.75% which was a reduction from the applicable margin of 3.00% prior to the First Amendment. The Second Amendment, dated September 16, 2023, provides for, among other things, the refinancing of the Company’s outstanding term loans B under the senior secured term loan facility due 2029 in an aggregate principal amount of $3.01 billion with a new tranche of term loans B in an aggregate principal amount of $3.01 billion. Under the Second Amendment, the outstanding term loans B under the amended senior secured term loan facility due 2029 will bear interest, at a rate per annum equal to the SOFR plus an applicable margin of 2.50% which is a reduction from the applicable margin of 2.75% provided for in the First Amendment. See Note 11 to our consolidated financial statements for further discussion. During the fiscal year 2023, the Company repaid $1.43 billion of the outstanding borrowings under the New Tranche B Term Loan and $175.5 million in full repayment of all outstanding borrowings under the bridge credit facility. Through January 5, 2024, the Company was in compliance with all applicable financial covenants included in the terms of its debt arrangements. The Company has commitments under the Revolving Facility of $747.5 million. The Revolving Facility bears interest at a rate per annum equal to, at the Company’s option, either a base rate (such as prime rate) or SOFR, plus, in each case, an applicable margin. During the twelve months ended January 5, 2024, there were no borrowings under this Revolving Facility and no balance was outstanding at January 5, 2024. The Company also has a line of credit with one bank that provides for borrowings of Japanese yen for the Company’s Japanese subsidiary equivalent to an aggregate of approximately $9.23 million. There were no outstanding borrowings under this line of credit and no balance was outstanding at January 5, 2024.

Cash and cash requirements

Our cash and cash equivalents include cash on hand and highly liquid debt securities with original maturities of three months or less, which are valued at cost and approximate fair value. We utilize a variety of funding strategies in an effort to ensure that our worldwide cash is available in the locations in which it is needed. Our restricted cash represented cash held in a “Rabbi” trust and is not available for general corporate purposes. See Note 6 to the consolidated financial statements for additional information. The Company had no restricted cash as of January 5, 2024.

Cash requirements

We have cash requirements to support working capital needs, capital expenditures, business acquisitions, contractual obligations, commitments, principal and interest payments on debt and other liquidity requirements associated with our operations. We generally intend to use available cash and funds generated from our operations to meet these cash requirements, but in the event that additional liquidity is required we may also borrow under our Revolving Facility. The following table summarizes our short and long-term cash requirements as of January 5, 2024:

Long-term debt and interest payments on long-term debt. We have contractual obligations for principal and interest payments on our long-term debt. See Note 11 of the consolidated financials for additional information. Debt obligations are classified based on their stated maturity date, regardless of their classification on the Company's consolidated balance sheets. Interest projections on both variable and fixed rate long-term debt are based on interest rates effective as of January 5, 2024 and do not include \$119.08 million for net unamortized discounts and debt issuance costs. On August 2, 2022, the Company entered into a floating-to-fixed interest rate swap agreement to hedge the variability in SOFR-based interest payments associated with \$2.54 billion of its \$3.24 billion Initial Term Loan Facility. The notional amount of the swap is \$1.82 billion at January 5, 2024 and is scheduled to decrease quarterly and will expire on January 4, 2026. The impact of the interest rate swap is not considered in the interest payments above. Capital purchase obligations. We have capital purchase obligations that represent commitments for the construction or purchase of property, plant and equipment. They were not recorded as liabilities on the Company's consolidated balance sheet as of January 5, 2024, as the Company had not yet received the related goods or taken title to the property. We expect capital expenditure spending to be approximately \$455 million in 2024 for growth capacity investments and the construction of our new manufacturing facility in Nevada. Supply purchase obligations. We have non-cancelable commitments, including take-or-pay contracts, that are not presented as capital purchase commitments above. They were not recorded as liabilities on the Company's consolidated balance sheet as of January 5, 2024, as the Company had not yet received the related goods or taken title to the property. Operating and financing lease commitments. Commitments under operating and financing leases primarily relate to leasehold properties. See Note 15 of the consolidated financials for additional information. Income tax liabilities. Of the tax liabilities included in the table above, \$88.01 million relates to uncertain tax positions. We are unable to accurately predict when these amounts will be realized or released. However, it is reasonably possible that there could be significant changes to our unrecognized tax benefits in the next twelve months due to an unforeseeable event (such as a tax audit settlement). See Note 17 of the consolidated financials for additional information.

New Accounting Pronouncements

Recently adopted accounting pronouncements Refer to Note 1 to the Company's consolidated financial statements for a discussion of accounting pronouncements implemented in 2023. Recently issued accounting pronouncements Refer to Note 1 of the Company's consolidated financial statements for a discussion of accounting pronouncements recently issued but not yet adopted. Non-generally accepted accounting principles Information The Company's consolidated financial statements are prepared in conformity with generally accepted accounting principles. The Company also utilizes certain non-generally accepted accounting principles financial measures as a complement to financial measures provided in accordance with generally accepted accounting principles in order to better assess and reflect trends affecting the Company's business and results of operations. These non-generally accepted accounting principles financial measures include Adjusted EBITDA and Adjusted Operating Income, together with related measures thereof, and Non-generally accepted accounting principles EPS, as well as certain other supplemental non-generally accepted accounting principles financial measures included in the discussion of the Company's financial results. Adjusted EBITDA is defined by the Company as net income before, as applicable, (1) income tax (benefit) expense, (2) interest expense, (3) interest income, (4) other expense, net, (5) goodwill impairment, (6) deal and transaction costs, (7) integration costs, (8) contractual costs and non-cash integration costs, (9) restructuring costs, (10) loss (gain) on sale of businesses, (11) charge for fair value write-up of acquired inventory sold, (12) gain on termination of the Alliance Agreement, (13) impairment of long-lived assets, (14) amortization of intangible assets and

(15) depreciation. Adjusted Operating Income is defined by the Company as Adjusted EBITDA exclusive of the depreciation addback noted above. The Company also utilizes ratios of non-generally accepted accounting principles financial measures such as Adjusted EBITDA to Company net sales and Adjusted Operating Income (referred to as Adjusted EBITDA Margin and Adjusted Operating Margin, respectively). Non- generally accepted accounting principles Net Income is defined by the Company as net income before, as applicable, (1) goodwill impairment, (2) deal and transaction costs, (3) integration costs, (4) contractual costs and non-cash integration costs, (5) restructuring costs, (6) loss on extinguishment of debt and modification, (7) loss (gain) on sale of businesses, (8) gain on termination of the Alliance Agreement, (9) Lubrex Chemicals termination fee, net, (10) charge for fair value write-up of sale of acquired inventory, (11) interest expense, net, (12) impairment of long-lived assets, (13) amortization of intangible assets, (14) the tax effect of the foregoing adjustments to net income, stated on a per share basis, divided by diluted weighted average shares outstanding. Non-generally accepted accounting principles EPS is defined as our Non-generally accepted accounting principles Net Income divided by our diluted weighted-average shares outstanding. The Company provides supplemental non-generally accepted accounting principles financial measures to help management and investors to better understand its business and believes these measures provide investors and analysts additional and meaningful information for the assessment of the Company's ongoing results. Management also uses these non-generally accepted accounting principles measures to assist in the evaluation of the performance of its business segments and to make operating decisions. Management believes the Company's non-generally accepted accounting principles measures help indicate the Company's baseline performance before certain gains, losses or other charges that may not be indicative of the Company's business or future outlook and offer a useful view of business performance in that the measures provide a more consistent means of comparing performance. The Company believes the non-generally accepted accounting principles measures aid investors' overall understanding of the Company's results by providing a higher degree of transparency for such items and providing a level of disclosure that will help investors understand how management plans, measures and evaluates the Company's business performance. Management believes that the inclusion of non-generally accepted accounting principles measures provides greater consistency in its financial reporting from period-to-period and facilitates investors' understanding of the Company's historical operating trends by providing an additional basis for comparisons to prior periods. Management uses Adjusted EBITDA and Adjusted Operating Income to assist it in evaluations of the Company's operating performance by excluding items that management does not consider as relevant in the results of its ongoing operations. Internally, these non-generally accepted accounting principles measures are used by management for planning and forecasting purposes, including the preparation of internal budgets; for allocating resources to enhance financial performance; for evaluating the effectiveness of operational strategies; and for evaluating the Company's capacity to fund capital expenditures, secure financing and expand its business. In addition, and as a consequence of the importance of these non-generally accepted accounting principles financial measures in managing its business, the Company's Board of Directors uses non-generally accepted accounting principles financial measures in the evaluation process to determine management compensation. The Company believes that certain analysts and investors use Adjusted EBITDA, Adjusted Operating Income and Non-generally accepted accounting principles EPS as supplemental measures to evaluate the overall operating performance of firms in the Company's industry. Additionally, lenders or potential lenders use Adjusted EBITDA measures to evaluate the Company's creditworthiness. The presentation of non-generally accepted accounting principles financial measures is not meant to be considered in isolation, as a substitute for, or superior to, financial measures or information provided in accordance with generally accepted accounting principles. Management strongly encourages investors to review the Company's consolidated financial statements in their entirety and to not rely on any single financial measure. Management notes that the use of non-generally accepted accounting principles measures has limitations, including but not limited to: First, non-generally accepted accounting principles financial measures are not standardized. Accordingly, the methodology used to produce the Company's non-generally accepted accounting principles financial measures is not computed under generally accepted accounting principles and may differ notably from the methodology used by other companies. For example, the Company's non-generally accepted

accounting principles measure of Adjusted EBITDA may not be directly comparable to EBITDA or an Adjusted EBITDA measure reported by other companies. Second, the Company's non-generally accepted accounting principles financial measures exclude items such as amortization and depreciation that are recurring. Amortization of intangibles and depreciation have been, and will continue to be for the foreseeable future, significant recurring expenses with an impact upon the Company's results of operations, notwithstanding the lack of immediate impact upon cash flows. Third, there is no assurance that the Company will not have future charges for fair value write-up of acquired inventory, restructuring activities, deal and transaction costs, integration costs, asset or goodwill impairments, loss on extinguishment of debt or similar items and, therefore, may need to record additional charges (or credits) associated with such items, including the tax effects thereon. The exclusion of these items in the Company's non-generally accepted accounting principles measures should not be construed as an implication that these costs are unusual, infrequent or non-recurring. Management considers these limitations by providing specific information regarding the generally accepted accounting principles amounts excluded from these non-generally accepted accounting principles financial measures and evaluating these non-generally accepted accounting principles financial measures together with their most directly comparable financial measures calculated in accordance with generally accepted accounting principles. The calculations of Adjusted EBITDA, Adjusted Operating Income, and Non-generally accepted accounting principles EPS, and reconciliations between these financial measures and their most directly comparable generally accepted accounting principles equivalents, are presented below in the accompanying tables.

The reconciliation of generally accepted accounting principles measures to adjusted operating income and Adjusted EBITDA for the years ended January 5, 2024 and 2022 are presented below:

1 Non-cash impairment charges associated with goodwill.2 Deal and transaction costs associated with Quantum Materials Division acquisition and completed and announced divestitures.3 Represents professional and vendor fees recorded in connection with services provided by consultants, accountants, lawyers and other third-party service providers to assist us in integrating Quantum Materials Division into our operations. These fees arise outside of the ordinary course of our continuing operations.4 Represents severance charges related to the integration of the Quantum Materials Division acquisition.5 Represents retention charges related directly to the Quantum Materials Division acquisition and completed and announced divestitures, and are not part of our normal, recurring cash operating expenses. 6 Represents other employee related costs and other costs incurred relating to the Quantum Materials Division acquisition and the completed and announced divestitures. These costs arise outside of the ordinary course of our continuing operations.7 Represents non-recurring costs associated with the Quantum Materials Division retention program that was agreed upon and set forth in the definitive acquisition agreement.8 Represents the non-cash incremental expense associated with adopting retirement vesting obligations on Zava Technologies equity awards, similar to those of Quantum Materials Division equity awards.9 Relates to the change in control agreements that were in place with management of Quantum Materials Division prior to the acquisition and the associated expense post-acquisition. 10 Restructuring charges resulting from cost saving initiatives.11 Loss (gain) from the sale of our businesses.12 Represents the additional cost of goods sold recognized in connection with the step-up of inventory valuation related to Quantum Materials Division acquisition. 13 Gain on termination of the Alliance Agreement with Veralux Industrial Chemistry.14 Impairment of long-lived assets.15 Non-cash amortization expense associated with intangibles acquired in acquisitions.

The reconciliation of generally accepted accounting principles measures to Non-generally accepted accounting principles EPS for the years ended January 5, 2024 and 2022 are presented below:

1 Non-cash impairment charges associated with goodwill.2 Deal and transaction costs associated with the Quantum Materials Division acquisition and completed and announced divestitures.3 Represents professional and vendor fees recorded in connection with services provided by consultants, accountants, lawyers and other third-party service providers to assist us in integrating Quantum Materials Division into our operations. These fees arise outside of the ordinary course of our continuing operations.4

Represents severance charges related to the integration of Quantum Materials Division.5 Represents retention charges related directly to the Quantum Materials Division acquisition and completed and announced divestitures, and are not part of our normal, recurring cash operating expenses. 6 Represents other employee-related costs and other costs incurred relating to the Quantum Materials Division acquisition and completed and announced divestitures. These costs arise outside of the ordinary course of our continuing operations.7 Represents non-recurring costs associated with the QUANTUM retention program that was agreed upon and set forth in the definitive acquisition agreement.8 Represents the non-cash incremental expense associated with adopting retirement vesting obligations on Zava Technologies equity awards, similar to those of QUANTUM Materials equity awards.9 Relates to the change in control agreements that were in place with management of Quantum Materials Division prior to the acquisition and the associated expense post-acquisition. 10 Restructuring charges resulting from cost saving initiatives.11 Non-recurring loss on extinguishment of debt and modification of our debt.12 Loss (gain) from the sale of our businesses.13 Gain on termination of the Alliance Agreement with Veralux Industrial Chemistry.14 Non-recurring gain from the termination fee with Lubrex Chemicals. 15 Represents the additional cost of goods sold recognized in connection with the step-up of inventory valuation related to the Quantum Materials Division acquisition. 16 Non-recurring interest costs related to the financing of the Quantum Materials Division acquisition. 17 Impairment of long-lived assets. 18 Non-cash amortization expense associated with intangibles acquired in acquisitions.19 The tax effect of pre-tax adjustments to net income was calculated using the applicable marginal tax rate for each respective year.